Drive Reduction Theory (C. Hull)

Overview:

Hull developed a version of behaviorism in which the stimulus (S) affects the organism (O) and the resulting response (R) depends upon characteristics of both O and S. In other words, Hull was interested in studying intervening variables that affected behavior such as initial drive, incentives, inhibitors, and prior training (habit strength). Like other forms of behavior theory, reinforcement is the primary factor that determines learning. However, in Hull's theory, drive reduction or need satisfaction plays a much more important role in behavior than in other frameworks (i.e., Thorndike, Skinner) .

Hull's theoretical framework consisted of many postulates stated in mathematical form; They include: (1) organisms possess a hierarchy of needs which are aroused under conditions of stimulation and drive, (2) habit strength increases with activities that are associated with primary or secondary reinforcement, (3) habit strength aroused by a stimulus other than the one originally conditioned depends upon the closeness of the second stimulus in terms of discrimination thresholds, (4) stimuli associated with the cessation of a response become conditioned inhibitors, (5) the more the effective reaction potential exceeds the reaction theshold, the shorter the latency of response. As these postulates indicate, Hull proposed many types of variables that accounted for generalization, motivation, and variability (oscillation) in learning.

One of the most important concepts in Hull's theory was the habit strength hierarchy: for a given stimulus, an organism can respond in a number of ways. The likelihood of a specific response has a probability which can be changed by reward and is affected by various other variables (e.g. inhibition). In some respects, habit strength hierarchies resemble components of cognitive theories such as schema and production systems .

Scope/Application:

Hull's theory is meant to be a general theory of learning. Most of the research underlying the theory was done with animals, except for Hull et al. (1940) which focused on verbal learning. Miller & Dollard (1941) represents an attempt to apply the theory to a broader range of learning phenomena. As an interesting aside, Hull began his career researching hypnosis – an area that landed him in some controversy at Yale (Hull, 1933).

Example:

Here is an example described by Miller & Dollard (1941): A six year old girl who is hungry and wants candy is told that there is candy hidden under one of the books in a bookcase. The girl begins to pull out books in a random manner until she finally finds the correct book (210 seconds). She is sent out of the room and a new piece of candy is hidden under the same book. In her next search, she is much more directed and finds the candy in 86 seconds. By the ninth repetition of this experiment, the girl finds the candy immediately (2 seconds). The girl exhibited a drive for the candy and looking under books represented her responses to reduce this drive. When she eventually found the correct book, this particular response was rewarded, forming a habit. On subsequent trials, the strength of this habit was increased until it became a single stimulus-response connection in this setting.

Principles:

1. Drive is essential in order for responses to occur (i.e., the student must want to learn).

2. Stimuli and responses must be detected by the organism in order for conditioning to occur (i.e., the student must be attentive).

3. Response must be made in order for conditioning to occur (i.e., the student must be active).

4. Conditioning only occurs if the reinforcement satisfied a need (i.e, the learning must satisfy the learner's wants).

Connectionism (E. Thorndike)

Overview:

The learning theory of Thorndike represents the original S-R framework of behavioral psychology: Learning is the result of associations forming between stimuli and responses. Such associations or "habits" become strengthened or weakened by the nature and frequency of the S-R pairings. The paradigm for S-R theory was trial and error learning in which certain responses come to dominate others due to rewards. The hallmark of connectionism (like all behavioral theory) was that learning could be adequately explained without refering to any unobservable internal states.

Thorndike's theory consists of three primary laws: (1) law of effect - responses to a situation which are followed by a rewarding state of affairs will be strengthened and become habitual responses to that situation, (2) law of readiness - a series of responses can be chained together to satisfy some goal which will result in annoyance if blocked, and (3) law of exercise - connections become strengthened with practice and weakened when practice is discontinued. A corollary of the law of effect was that responses that reduce the likelihood of achieving a rewarding state (i.e., punishments, failures) will decrease in strength.

The theory suggests that transfer of learning depends upon the presence of identical elements in the original and new learning situations; i.e., transfer is always specific, never general. In later versions of the theory, the concept of "belongingness" was introduced; connections are more readily established if the person perceives that stimuli or responses go together (c.f. Gestalt principles). Another concept introduced was "polarity" which specifies that connections occur more easily in the direction in which they were originally formed than the opposite. Thorndike also introduced the "spread of effect" idea, i.e., rewards affect not only the connection that produced them but temporally adjacent connections as well.

Scope/Application:

Connectionism was meant to be a general theory of learning for animals and humans. Thorndike was especially interested in the application of his theory to education including mathematics (Thorndike, 1922), spelling and reading (Thorndike, 1921), measurement of intelligence (Thorndike et al., 1927) and adult learning (Thorndike at al., 1928).

Example:

The classic example of Thorndike's S-R theory was a cat learning to escape from a "puzzle box" by pressing a lever inside the box. After much trial and error behavior, the cat learns to associate pressing the lever (S) with opening the door (R). This S-R connection is established because it results in a satisfying state of affairs (escape from the box). The law of exercise specifies that the connection was established because the S-R pairing occurred many times (the law of effect) and was rewarded (law of effect) as well as forming a single sequence (law of readiness).

Principles:

1. Learning requires both practice and rewards (laws of effect /exercise)

2. A series of S-R connections can be chained together if they belong to the same action sequence (law of readiness).

3. Transfer of learning occurs because of previously encountered situations.

4. Intelligence is a function of the number of connections learned.

Operant Conditioning (B.F. Skinner)

Overview:

The theory of B.F. Skinner is based upon the idea that learning is a function of change in overt behavior. Changes in behavior are the result of an individual's response to events (stimuli) that occur in the environment. A response produces a consequence such as defining a word, hitting a ball, or solving a math problem. When a particular Stimulus-Response (S-R) pattern is reinforced (rewarded), the individual is conditioned to respond. The distinctive characteristic of operant conditioning relative to previous forms of behaviorism (e.g., Thorndike, Hull) is that the organism can emit responses instead of only eliciting response due to an external stimulus.

Reinforcement is the key element in Skinner's S-R theory. A reinforcer is anything that strengthens the desired response. It could be verbal praise, a good grade or a feeling of increased accomplishment or satisfaction. The theory also covers negative reinforcers -- any stimulus that results in the increased frequency of a response when it is withdrawn (different from adversive stimuli -- punishment -- which result in reduced responses). A great deal of attention was given to schedules of reinforcement (e.g. interval versus ratio) and their effects on establishing and maintaining behavior.

One of the distinctive aspects of Skinner's theory is that it attempted to provide behavioral explanations for a broad range of cognitive phenomena. For example, Skinner explained drive (motivation) in terms of deprivation and reinforcement schedules. Skinner (1957) tried to account for verbal learning and language within the operant conditioning paradigm, although this effort was strongly rejected by linguists and psycholinguists. Skinner (1971) deals with the issue of free will and social control.

Scope/Application:

Operant conditioning has been widely applied in clinical settings (i.e., behavior modification) as well as teaching (i.e., classroom management) and instructional development (e.g., programmed instruction). Parenthetically, it should be noted that Skinner rejected the idea of theories of learning (see Skinner, 1950).

Example:

By way of example, consider the implications of reinforcement theory as applied to the development of programmed instruction (Markle, 1969; Skinner, 1968)

1. Practice should take the form of question (stimulus) - answer (response) frames which expose the student to the subject in gradual steps

2. Require that the learner make a response for every frame and receive immediate feedback

3. Try to arrange the difficulty of the questions so the response is always correct and hence a positive reinforcement

4. Ensure that good performance in the lesson is paired with secondary reinforcers such as verbal praise, prizes and good grades.

Principles:

1. Behavior that is positively reinforced will reoccur; intermittent reinforcement is particularly effective

2. Information should be presented in small amounts so that responses can be reinforced ("shaping")

3. Reinforcements will generalize across similar stimuli ("stimulus generalization") producing secondary conditioning

Carl Rogers Theories Of Personality

Personality theories have provided a wide range of information on the behavior of an individual and what constitutes him.

Carl Roger was a clinical psychologist. His outlook on human behavior was it is “exquisitely rational”. According to him, the inner nature of man is actually positive and he is a trustworthy individual.
His theory is a very valuable contribution to the study of freedom, importance of the self, study of person and recognizing agency. They were rich and matured.

Rogers complete theory was built on one single “force of life” i.e ‘the self actualization tendency’.

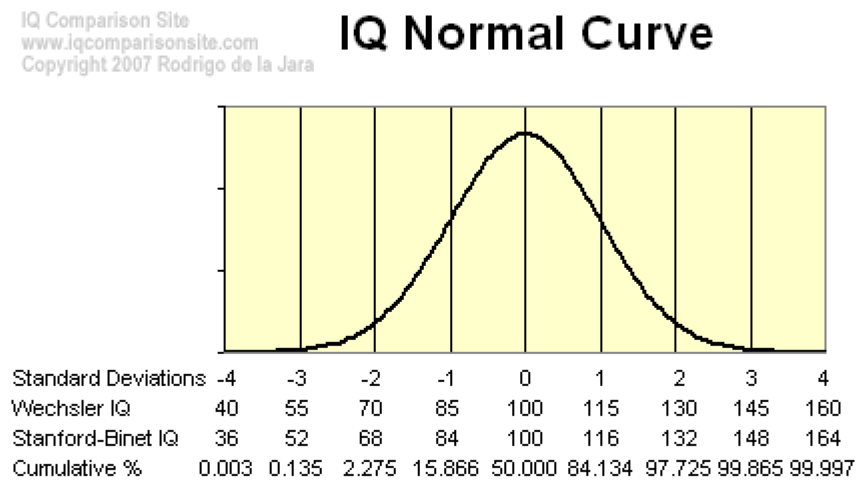
Actualizing Tendency
According to Roger, every individual has a hidden actualizing tendency. This tendency is constructive, directional and is present in every living being. It can be held back but can never be killed until the individual is destroyed. Roger says that every individual strives hard to make the best of his existence.

Self
‘Self’ is the main concept in Rogers theory. It is involves awareness of being and functioning and establishes through interaction with other individuals.

Self-Actualizing Tendency
It is the actualization of that part of experience which is symbolized in the self. In brief, self actualization is a master motive- It is the development of the psychology which can only be achieved when all the basic and mental needs are met.

Carl Roger was interested in improving the human conditions. His person-centered therapy is the best contribution to psychology. He always treated people ethnically and encouraged human. According to him, psychology is a ‘Human Science’ rather than a natural science.

IQ Basics



Graph drawn in Excel using the NORMDIST function.

This is a quick explanation of IQ, put up due to popular demand. There are many books on psychology or intelligence that would provide a more rigorous explanation of IQ.

What is intelligence? The definition I like is that intelligence is "the ability to learn or understand or to deal with new or trying situations ... also : the skilled use of reason" (7)*. I have heard some people misuse the word smart to mean knowledgeable. That is like confusing velocity with distance. That one can lead to the other does not mean that they are the same thing.

I.Q. = Intelligence Quotient

Originally, "IQ" tests were created to be able to identify children who might need special education due to their retarded mental development (1). Binet's test included varied questions and tasks. The tasks even included unwrapping a piece of candy and comparing the weights of different objects (4)!

To relate the mental development of a child to the child's chronological age the IQ was invented. IQ = (MA/CA) * 100. The intelligence quotient was equal to 100 times the Mental Age divided by the Chronological Age. For example, if a certain child started reading, etc., at the age of 3 (CA) and average children start reading, etc., at the age of 6 (MA), the child would get an IQ score of 200. (Such a score is very, very rare). Since people wanted to also use IQs for adults, that formula was not very useful since raw scores start to level off around the age of 16 (2).

Thus the deviation IQ replaced the ratio IQ. It compares people of the same age or age category and assumes that IQ is normally distributed, that the average (mean) is 100 and that the standard deviation is something like 15 (IQ tests sometimes differ in their standard deviations).

What is a standard deviation (SD)? Simply put, the standard deviation is a measure of the spread of the sample from the mean. As a rule of thumb, about 2/3 of a sample is within 1 standard deviation from the mean. About 95% of the sample will be within 2 standard deviations from the mean (3).

With the standard deviation and a mean, you can calculate percentiles. Percentiles tell you the percent of people that have a score equal to or lower than a certain score.

High IQ societies ask for certain percentile scores on IQ tests for you to be eligible to join them. Mensa asks for scores at the 98th percentile or higher. For a list of the selection criteria of other societies, click here.

There have been various classification systems for IQ.

Terman's classification was (6):

IQ Range	**Classification**
140 and over	Genius or near genius
120-140	Very superior intelligence
110-120	Superior intelligence
90-110	Normal or average intelligence
80-90	Dullness
70-80	Borderline deficiency
Below 70	Definite feeble-mindedness

(Terman wrote the Stanford-Binet test (1), which has a SD of 16.)

Later, Wechsler thought that it would be much more legitimate to base his classifications on the Probable Error (PE) so his classification was (6):

Classification	**IQ Limits**	**Percent Included**
Very Superior	128 and over	2.2
Superior	120-127	6.7
Bright Normal	111-119	16.1
Average	91-110	50
Dull Normal	80-90	16.1
Borderline	66-79	6.7
Defective	65 and below	2.2

Mental deficiency used to be more finely classified using the following technical terms that later began to be abused by the rest of society (5):

IQ Range	**Classification**
70-80	Borderline deficiency
50-69	Moron
20-49	Imbecile
below 20	Idiot

These are now largely obsolete and mental deficiency is now generally called mental retardation. The following is the currently used classification of retardation in the USA (5):

IQ Range	**Classification**
50-69	Mild
35-49	Moderate
20-34	Severe
below 20	Profound

Moreover, "educable mentally retarded" is roughly equivalent to mild mental retardation, and "trainable" mentally retarded is roughly equivalent to moderate (5). The DSM now requires an assessment of a person's adaptive functioning as an additional criterion for labeling someone retarded. IQ is not enough. Maybe the same sort of thing should be done for labeling somebody a genius.

The **theory of multiple intelligences** was proposed by Howard Gardner in 1983 to analyze and better describe the concept of intelligence.

Gardner argues that the concept of intelligence as traditionally defined in psychometrics (IQ tests) does not sufficiently describe the wide variety of cognitive abilities humans display. For example, the theory states that a child who learns to multiply easily is not necessarily more intelligent than a child who has stronger skills in another *kind* of intelligence. The child who takes more time to master simple multiplication 1) may best learn to multiply through a different approach, 2) may excel in a field outside of mathematics, or 3) may even be looking at and understand the multiplication process at a fundamentally deeper level. Such a fundamentally deeper understanding can result in what looks like slowness and can hide a mathematical intelligence potentially higher than that of a child who quickly memorizes the multiplication table despite a less detailed understanding of the process of multiplication.

The theory has been met with mixed responses. Many psychologists feel that a differentiation of the concept of intelligence is not supported by empirical evidence, but many educationalists support the practical value of the approaches suggested by the theory.

Gardner has articulated eight basic types of intelligence to date, without claiming that this is a complete list.[1] Gardner's original list included seven of these; in 1999 he added a naturalist intelligence. He has also considered existential intelligence and moral intelligence, but does not find sufficient evidence for these based upon his articulated criteria,[2] which include:[3]

- the potential for brain isolation by brain damage,
- its place in evolutionary history,
- the presence of core operations,
- susceptibility to encoding (symbolic expression),
- a distinct developmental progression,
- the existence of idiot-savants, prodigies and other exceptional people,
- support from experimental psychology and psychometric findings.

The theory's nine currently accepted intelligences are: (Ref: *Educational Psychology*, Robert Slavin. 2009, 117)

- Spatial
- Linguistic
- Logical-mathematical
- Bodily-kinesthetic
- Musical
- Interpersonal

- Intrapersonal
- Naturalistic
- Existential

[edit] Spatial

This area deals with spatial judgment and the ability to visualize with the mind's eye. Careers which suit those with this type of intelligence include artists, designers and architects. A spatial person is also good with puzzles.

[edit] Linguistic

This area has to do with words, spoken or written. People with high verbal-linguistic intelligence display a facility with words and languages. They are typically good at reading, writing, telling stories and memorizing words along with dates. They tend to learn best by reading, taking notes, listening to lectures, and by discussing and debating about what they have learned. Those with verbal-linguistic intelligence learn foreign languages very easily as they have high verbal memory and recall, and an ability to understand and manipulate syntax and structure.

Careers that suit those with this intelligence include writers, lawyers, policemen, philosophers, journalists, politicians, poets and teachers.[citation needed]

[edit] Logical-mathematical

This area has to do with logic, abstractions, reasoning and numbers. While it is often assumed that those with this intelligence naturally excel in mathematics, chess, computer programming and other logical or numerical activities, a more accurate definition places less emphasis on traditional mathematical ability and more on reasoning capabilities, recognising abstract patterns, scientific thinking and investigation and the ability to perform complex calculations. It correlates strongly with traditional concepts of "intelligence" or IQ.

Careers which suit those with this intelligence include scientists, physicists, mathematicians, logicians, pilots, engineers, doctors, economists and philosophers.[citation needed]

[edit] Bodily-kinesthetic

The core elements of the bodily-kinesthetic intelligence are control of one's bodily motions and the capacity to handle objects skillfully (206). Gardner elaborates to say that this intelligence also includes a sense of timing, a clear sense of the goal of a physical action, along with the ability to train responses so they become like reflexes.

In theory, people who have bodily-kinesthetic intelligence should learn better by involving muscular movement (e.g. getting up and moving around into the learning experience), and are generally good at physical activities such as sports or dance. They may enjoy acting or performing, and in general they are good at building and making things. They often learn best by doing something physically, rather than by reading or hearing about it. Those with strong bodily-kinesthetic intelligence seem to use what might be termed muscle memory - they remember things through their body such as verbal memory.

Careers that suit those with this intelligence include: athletes, pilots, dancers, musicians, actors, surgeons, doctors, builders, police officers, and soldiers. Although these careers can be duplicated through virtual simulation, they will not produce the actual physical learning that is needed in this intelligence.[4]

[edit] Musical

This area has to do with sensitivity to sounds, rhythms, tones, and music. People with a high musical intelligence normally have good pitch and may even have absolute pitch, and are able to sing, play musical instruments, and compose music. Since there is a strong auditory component to this intelligence, those who are strongest in it may learn best via lecture. Language skills are typically highly developed in those whose base intelligence is musical. In addition, they will sometimes use songs or rhythms to learn. They have sensitivity to rhythm, pitch, meter, tone, melody or timbre.

Careers that suit those with this intelligence include instrumentalists, singers, conductors, disc-jockeys, orators, writers and composers.

[edit] Interpersonal

This area has to do with interaction with others. In theory, people who have a high interpersonal intelligence tend to be extroverts, characterized by their sensitivity to others' moods, feelings, temperaments and motivations, and their ability to cooperate in order to work as part of a group. They communicate effectively and empathize easily with others, and may be either leaders or followers. They typically learn best by working with others and often enjoy discussion and debate.

Careers that suit those with this intelligence include sales, politicians, managers, teachers and social workers.[5]

[edit] Intrapersonal

This area has to do with introspective and self-reflective capacities. People with intrapersonal intelligence are intuitive and typically introverted. They are skillful at deciphering their own feelings and motivations. This refers to having a deep understanding of the self; what are your strengths/ weaknesses, what makes you unique, you can predict your own reactions/ emotions.

Careers which suit those with this intelligence include philosophers, psychologists, theologians, lawyers and writers. People with intrapersonal intelligence also prefer to work alone.

[edit] Naturalistic

This area has to do with nature, nurturing and relating information to one’s natural surroundings. Careers which suit those with this intelligence include naturalists, farmers and gardeners.

[edit] Existential

Ability to contemplate phenomena or questions beyond sensory data, such as the infinite and infinitesimal. Careers which suit those with this intelligence include mathematicians, physicists, scientists, cosmologists and philosophers.

[edit] Use in education

Traditionally, schools have emphasized the development of logical intelligence and linguistic intelligence (mainly reading and writing). IQ tests (given to about 1,000,000 students each year) focus mostly on logical and linguistic intelligence as well. While many students function well in this environment, there are those who do not. Gardner's theory argues that students will be better served by a broader vision of education, wherein teachers use different methodologies, exercises and activities to reach all students, not just those who excel at linguistic and logical intelligence.

Many teachers see the theory as simple common sense. Some say that it validates what they already know: that students learn in different ways. On the other hand, James Traub's article in *The New Republic* notes that Gardner's system has not been accepted by most academics in intelligence or teaching.

George Miller, the esteemed psychologist credited with discovering the mechanisms by which short term memory operates, wrote in *The New York Times Book Review* that Gardner's argument boiled down to "hunch and opinion" (p. 20). Gardner's subsequent work has done very little to shift the balance of opinion. A recent issue of *Psychology, Public Policy, and Law* devoted to the study of intelligence contained virtually no reference to Gardner's work. Most people who study intelligence view M.I. theory as rhetoric rather than science, and they're divided on the virtues of the rhetoric.

The application of the theory of multiple intelligences varies widely. It runs the gamut from a teacher who, when confronted with a student having difficulties, uses a different approach to teach the material, to an entire school using MI as a framework. In general, those who subscribe to the theory strive to provide opportunities for their students to use and develop all the different intelligences, not just the few at which they naturally excel.

A Harvard-led study of 41 schools using the theory came to the conclusion that in these schools there was "a culture of hard work, respect, and caring; a faculty that collaborated and learned from each other; classrooms that engaged students through constrained but meaningful choices, and a sharp focus on enabling students to produce high-quality work."[6]

Of the schools implementing Gardner's theory, the most well-known is New City School, in St. Louis, Missouri, which has been using the theory since 1988. The school's teachers have produced two books for teachers, *Celebrating Multiple Intelligences* and *Succeeding With Multiple Intelligences* and the principal, Thomas Hoerr, has written *Becoming a Multiple Intelligences School* as well as many articles on the practical applications of the theory. The school has also hosted four conferences, each attracting over 200 educators from around the world and remains a valuable resource for teachers interested in implementing the theory in their own classrooms.

Thomas Armstrong argues that Waldorf education organically engages all of Gardner's original seven intelligences.[7]

[edit] Questions

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Questions raised about Gardner's theory include:

- What kind of correlations exist between the intelligences, or are they completely independent?
- Should schools be focusing on teaching to students' strengths or on remediating where they are weak?
- To what extent should students be aware of their profile in the various intelligences?

- **Behaviorism** (or **behaviourism**), also called the **learning perspective** (where any physical action is a behavior), is a philosophy of psychology based on the proposition that all things that organisms do—including acting, thinking and feeling—can and should be regarded as behaviors.[1] The behaviorist school of thought maintains that behaviors as such can be described scientifically without recourse either to internal physiological events or to hypothetical constructs such as the mind.[2] Behaviorism comprises the position that all theories should have observational correlates but that there are no philosophical differences between publicly observable processes (such as actions) and privately observable processes (such as thinking and feeling).[3]
- From early psychology in the 19th century, the behaviorist school of thought ran concurrently and shared commonalities with the psychoanalytic and Gestalt movements in psychology into the 20th century; but also differed from the mental philosophy of the Gestalt psychologists in critical ways.[*citation needed*] Its main influences were Ivan Pavlov, who investigated classical conditioning although he did not necessarily agree with Behaviorism or Behaviorists, Edward Lee Thorndike, John B. Watson who rejected introspective methods and sought to restrict psychology to experimental methods, and B.F. Skinner who conducted research on operant conditioning.[3]
- In the second half of the 20th century, behaviorism was largely eclipsed as a result of the cognitive revolution.[4][5] While behaviorism and cognitive schools of psychological thought may not agree theoretically, they have complemented each other in practical therapeutic applications, such as in cognitive–behavioral therapy that has demonstrable utility in treating certain pathologies, such as simple phobias, PTSD, and addiction. In addition, behaviorism sought to create a comprehensive model of the stream of behavior from the birth of the human to his death (see Behavior analysis of child development).

Versions

There is no classification generally agreed upon, but some titles given to the various branches of behaviorism include:

- Methodological: The behaviorism of Watson; the objective study of behavior; no mental life, no internal states; thought is covert speech.
- Radical: Skinner's behaviorism; is considered radical since it expands behavioral principles to processes within the organism; in contrast to methodological behaviorism; not mechanistic or reductionistic; hypothetical (mentalistic) internal states are not considered causes of behavior, phenomena must be observable at least to the individual experiencing them. Willard Van Orman Quine used many of radical behaviorism's ideas in his study of knowing and language.
- Teleological: Post-Skinnerian, purposive, close to microeconomics. Focuses on objective observation as opposed to cognitive processes.
- Theoretical: Post-Skinnerian, accepts observable internal states ("within the skin" once meant "unobservable", but with modern technology we are not so constrained); dynamic, but eclectic in choice of theoretical structures, emphasizes parsimony.
- Biological: Post-Skinnerian, centered on perceptual and motor modules of behavior, theory of behavior systems.

- Psychological behaviorism: Arthur W. Staats' unifying approach to behaviorism and psychology. He merges psychological concepts like "personality" within a behavioral model like BBR Basic Behavioral Repertoires.

Two subtypes are:

- Hullian and post-Hullian: theoretical, group data, not dynamic, physiological;
- Purposive: Tolman's behavioristic anticipation of cognitive psychology

[edit] Definition

B.F. Skinner was influential in defining radical behaviorism, a philosophy codifying the basis of his school of research (named the Experimental Analysis of Behavior, or EAB.) While EAB differs from other approaches to behavioral research on numerous methodological and theoretical points, radical behaviorism departs from methodological behaviorism most notably in accepting feelings, states of mind and introspection as existent and scientifically treatable. This is done by identifying them as something non-dualistic, and here Skinner takes a divide-and-conquer approach, with some instances being identified with bodily conditions or behavior, and others getting a more extended "analysis" in terms of behavior. However, radical behaviorism stops short of identifying feelings as causes of behavior.[1] Among other points of difference were a rejection of the reflex as a model of all behavior and a defense of a science of behavior complementary to but independent of physiology. Radical behaviorism has considerable overlap with other western philosophical positions such as American pragmatism.[6]

Relation to language

As Skinner turned from experimental work to concentrate on the philosophical underpinnings of a science of behavior, his attention turned to human language with *Verbal Behavior*[10] and other language-related publications;[11] *Verbal Behavior* laid out a vocabulary and theory for functional analysis of verbal behavior, and was strongly criticized in a review by Noam Chomsky.[12] Skinner did not respond in detail but claimed that Chomsky failed to understand his ideas,[13] and the disagreements between the two and the theories involved have been further discussed.[14][15] In addition; innate theory is opposed to behaviorist theory which claims that language is a set of habits that can be acquired by means of conditioning. According to some, this process that the behaviorists define is a very slow and gentle process to explain a phenomenon complicated as language learning. What was important for a behaviorist's analysis of human behavior was not language acquisition so much as the interaction between language and overt behavior. In an essay republished in his 1969 book *Contingencies of Reinforcement*,[16] Skinner took the view that humans could construct linguistic stimuli that would then acquire control over their behavior in the same way that external stimuli could. The possibility of such "instructional control" over behavior meant that contingencies of reinforcement would not always produce the same effects on human behavior as they reliably do in other animals. The focus of a radical behaviorist analysis of human behavior therefore shifted to an attempt to understand the interaction between instructional control and contingency control, and also to understand the behavioral processes that determine what instructions are constructed and what control they acquire over behavior. Recently a new, promising line of behavioral research on language was started under the name of Relational Frame Theory.

[edit] Molar versus molecular behaviorism

Skinner's view of behavior is most often characterized as a "molecular" view of behavior; that is, behavior can be decomposed into atomistic parts or molecules. This view is inconsistent with Skinner's complete description of behavior as delineated in other works, including his 1981 article "Selection by Consequences".[17] Skinner proposed that a complete account of behavior requires understanding of selection history at three levels: biology (the natural selection or phylogeny of the animal); behavior (the reinforcement history or ontogeny of the behavioral repertoire of the animal); and for some species, culture (the cultural practices of the social group to which the animal belongs). This whole organism then interacts with its environment. Molecular behaviorists use notions from melioration theory, negative power function discounting or additive versions of negative power function discounting.[18]

Molar behaviorists, such as Howard Rachlin, Richard Herrnstein, and William Baum, argue that behavior cannot be understood by focusing on events in the moment. That is, they argue that behavior is best understood as the ultimate product of an organism's history and that molecular behaviorists are committing a fallacy by inventing fictitious proximal causes for behavior. Molar behaviorists argue that standard molecular constructs, such as "associative strength", are better replaced by molar variables such as rate of reinforcement.[19] Thus, a molar behaviorist would describe "loving someone" as a pattern of loving behavior over time; there is no isolated, proximal cause of loving behavior, only a history of behaviors (of which the current behavior might be an example) that can be summarized as "love".

Psychoanalytic theory refers to the definition and dynamics of personality development which underlie and guide psychoanalytic and psychodynamic psychotherapy. First laid out by Sigmund Freud, psychoanalytic theory has undergone many refinements since his work (see psychoanalysis). Psychoanalytic theory came to full prominence as a critical force in the last third of the twentieth century as part of 'the flow of critical discourse after the 1960s'[1], and in association above all with the name of Jacques Lacan.

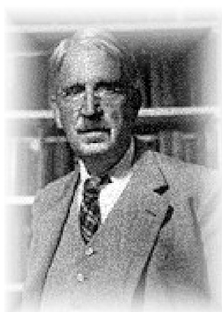
Psychoanalytic theory originated with the work of Sigmund Freud. Through his clinical work with patients suffering from mental illness, Freud came to believe that childhood experiences and unconscious desires influenced behavior. Based on his observations, he developed a theory that described development in terms of a series of psychosexual stages. According to Freud, conflicts that occur during each of these stages can have a lifelong influence on personality and behavior.

Psychoanalytic theory was an enormously influential force during the first half of the twentieth century. Those inspired and influenced by Freud went on to expand upon Freud's ideas and develop theories of their own. Of these neo-Freudians, Erik Erikson's ideas have become perhaps the best known. Erikson's eight-stage theory of psychosocial development describes growth and change throughout the lifespan, focusing on social interaction and conflicts that arise during different stages of development.

Introduction to John Dewey's Philosophy of Education

Education is life itself.
- John Dewey

John Dewey (1859-1952) believed that learning was active and schooling unnecessarily long and restrictive. His idea was that children came to school to do things and live in a community which gave them real, guided experiences which fostered their capacity to contribute to society. For example, Dewey believed that students should be involved in real-life tasks and challenges:



- maths could be learnt via learning proportions in cooking or figuring out how long it would take to get from one place to another by mule
- history could be learnt by experiencing how people lived, geography, what the climate was like, and how plants and animals grew, were important subjects

Dewey had a gift for suggesting activities that captured the center of what his classes were studying.

Dewey's education philosophy helped forward the "progressive education" movement, and spawned the development of "experiential education" programs and experiments.

Dewey's philosophy still lies very much at the heart of many bold educational experiments, such as Outward Bound. Read more about John Dewey, father of the experiential education movement.

Dewey is lauded as the greatest educational thinker of the 20th century. His theory of experience continues to be much read and discussed not only within education, but also in psychology and philosophy. Dewey's views continue to strongly influence the design of innovative educational approaches, such as in outdoor education, adult training, and experiential therapies.

In the 1920's / 1930's, John Dewey became famous for pointing out that the authoritarian, strict, pre-ordained knowledge approach of modern traditional education was too concerned with delivering knowledge, and not enough with understanding students' actual experiences.

Dewey became the champion, or philosophical father of experiential education, or as it was then referred to, progressive education. But he was also critical of completely "free, student-driven" education because students often don't know how to structure their own learning experiences for maximum benefit.

Why do so many students hate school? It seems an obvious, but ignored question.

Dewey said that an educator must take into account the unique differences between each student. Each person is different genetically and in terms of past experiences. Even when a standard curricula is presented using established pedagogical methods, each students will have a different quality of experience. Thus, teaching and curriculum must be designed in ways that allow for such individual differences.

For Dewey, education also a broader social purpose, which was to help people become more effective members of democratic society. Dewey argued that the one-way delivery style of authoritarian schooling does not provide a good model for life in democratic society. Instead, students need educational experiences which enable them to become valued, equal, and responsible members of society.

The most common misunderstanding about Dewey is that he was simply supporting progressive education. Progressive education, according to Dewey, was a wild swing in the philosophical pendulum, against traditional education methods. In progressive education, freedom was the rule, with students being relatively unconstrained by the educator. The problem with progressive education, said Dewey, is that freedom alone is no solution. Learning needs a structure and order, and must be based on a clear theory of experience, not simply the whim of teachers or students.

Thus, Dewey proposed that education be designed on the basis of a ***theory of experience***. We must understand the nature of how humans have the experiences they do, in order to design effective education. In this respect, Dewey's theory of experience rested on two central tenets -- continuity and interaction.

Continuity refers to the notion that humans are sensitive to (or are affected by) experience. Humans survive more by learning from experience after they are born than do many other animals who rely primarily on pre-wired instinct. In humans, education is critical for providing people with the skills to live in society. Dewey argued that we learn something from every experience, whether positive or negative and ones accumulated learned experience influences the nature of one's future experiences. Thus, every experience in some way influences all potential future experiences for an individual. Continuity refers to this idea that s each experience is stored and carried on into the future, whether one likes it or not.

Interaction builds upon the notion of continuity and explains how ***past experience interacts with the present situation***, to create one's present experience. Dewey's hypothesis is that your current experience can be understood as a function of your past (stored) experiences which interacting with the present situation to create an individual's experience. This explains the "one man's meat is another man's poison" maxim. Any situation can be experienced in profoundly different ways because of unique individual differences e.g., one student loves school, another hates the same school. This is important for educators to understand. Whilst they can't control students' past experiences, they can try to understand those past experiences so that better educational situations can be presented to the students. Ultimately, all a teacher has control over is the design of the present situation. The teacher with good insight into the effects of past experiences which students bring with them better enables the teacher to provide quality education which is relevant and meaningful for the students.

What is experiential learning?

Experiential learning has come to mean two different types of learning:
1. learning by yourself and
2. experiential education
[experiential learning through programs structured by others]
(Smith, 2003).

1. Experiential learning by yourself

Learning from experience by yourself might be called "nature's way of learning". It is "education that occurs as a direct participation in the events of life" (Houle, 1980, p. 221, quoted in Smith, 2003). It includes learning that comes about through reflection on everyday experiences. Experiential learning by yourself is also known as "informal education" and includes learning that is organised by learners themselves.

Related terms: Auto-didacticism, Self-teaching.

2. Experiential education

(Experiential learning through programs & activities structured by others)

Principles of experiential learning are used to design of *experiential education* programs. Emphasis is placed on the nature of participants' subjective experiences.

An experiential educator's role is to organize and facilitate *direct experiences of phenomenon* under the assumption that this will lead to genuine (meaningful and long-lasting) learning. This often also requires preparatory and reflective exercises.

Experiential education is often contrasted with *didactic education*, in which the teacher's role is to "give" information/knowledge to student and to prescribe study/learning exercises which have "information/knowledge transmission" as the main goal.

What is Experiential Education?

James Neill
Last updated:

- Students are actively involved
- Educators value students' experience
- Comes in many shapes & sizes
- According to Googlism
- Recommended reading

The mediocre teacher tells.
The good teacher explains.
The superior teacher demonstrates.
The great teacher inspires.
- William A. Ward

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Students are actively involved

In experiential education, the student becomes more actively involved in the learning process than than in traditional, didactic education. For example, going to a zoo and learning through observation and interaction with the zoo environment is experiential and in contrast to reading and talking about animals in a classroom. The main difference here, from a pedagogical point of view, is that the educator who takes his/her students to the zoo rather than stay in the classroom probably values direct experience more highly than abstract knowledge.

Educators' value the students' experience

Experiential education is based on experiential learning. Experiential educators operate under the assumption that:

educational goals can be effectively met by allowing the nature of learner's educational experience to influence the educational process

Experiential educators are generally aware that experiences alone are not inherently good for learning. Thus, experiential try to arrange particular sets of experiences which are conducive towards particular educational goals.

Experiential education comes in many shapes and sizes

Experiential education is widely implemented across a range of topics and mediums - for example, outdoor education, service learning, internships, and group-based learning projects. Many educational projects are experiential, but don't refer to themselves as such (e.g., excursions, physical education, manual arts, drama, art, and so on).

Empiricism

What is empiricism?

The word "empiricism" is derived from the Greek *empeiria*, the Latin translation of which is *experientia*, from which in turn we derive the word "experience." Empiricism also comes from *empiric* - a doctor who relies on practical experience. And in seventeenth- and eighteenth-century medicine, empiricism was synonymous with quackery, and in literary criticism the term is also generally employed to characterize an uninformed judgment.

Empiricism is a broad tradition in Western philosophy. The basic thesis of empiricism is that legitimate human knowledge arises from what is provided to the mind by the senses or by introspective awareness through experience.

Now to keep from defining one term by means of an almost equally ambiguous term, we should examine what we mean by experience. Different philosophers pick out different phenomena with the word: and even when they seem to pick out the same phenomenon, they may have different views as to the structure of the phenomenon that they call "experience." Aristotle took experience as the as yet unorganized product of sense perception and memory. This appears to be a common philosophical conception of the term. Memory is required so that what is perceived may be retained in the mind or remembered. When we say that we have learned something from experience we mean that we have come to know of it by the use of our senses. We have experience when we are sufficiently aware of what we have discovered in this way. Another connected sense of the term is the perception of feelings, sensations, and etc. as sense experiences. Awareness of these experiences is something that happens to us and it is in this sense passive. The statement that experience is the source of knowledge means that knowledge depends ultimately on the use of the senses and on what is discovered through them.

It seems an interesting parallel to note that just as the term "experience" is ultimately derived from the term "empiricism," empiricists maintain that all knowledge is ultimately derived from experience-sense experience.

Empiricism versus Rationalism

Empiricism is distinguished from the philosophical tradition of *rationalism*, which holds that human reason apart from experience is a basis for some kinds of knowledge. Knowledge attainable by reason alone, prior to experience, is called *a priori* knowledge; knowledge based upon experience is called *a posteriori* knowledge. For instance, "black cats are black" is an example of a priori knowledge. It is a tautology; its denial would be self-contradictory. "Desks are brown" is an example of a posteriori knowledge. It is not necessarily true unless all desks are by definition brown and to deny it would not be self-contradictory. We would refer to experience to settle the matter. These last statements are also referred to as *analytic* and *synthetic* statements respectively. Rationalists claim that knowledge can be derived from certain a priori truths by *deduction*. Empiricists claim that for human beings there is no pure reason and that all knowledge is a posteriori and derived from sense experience by *induction*.

On the side of rationalism are philosophers such as Plato, Descartes, Leibnez, and Spinoza to name a few. Plato, perhaps the most well known, profoundly distrusted the senses as a source of knowledge. He argued that knowledge can have as its object only that which is changeless, and since the physical world is ever-changing, one cannot have knowledge of it. He maintained that there is a changeless and perfect nonphysical world of "Forms," a world of concepts or properties like whiteness, justice, and beauty. Only reason can provide knowledge of this world of Forms; one cannot be aware of it by using one's senses. Next we turn to the Empiricists.

Who are the Empiricists?

Among the ancient philosophers, the Sophists were empiricists. Aristotle (384-322 BC) is sometimes said to be the founder of the empiricist tradition, although there are important rationalistic elements in his philosophy. Certainly Aquinas seemed to believe that he had Aristotle's authority for the view that there is nothing in the intellect which was not previously in the senses. Aristotle's place in the development of empiricism remains unclear, however.

Epicurus (341-270 BC) was a Greek philosopher who founded the system known as Epicureanism. Epicureans based their theory of knowledge on sense perception, asserting that sensations are invariably good evidence of their causes. They worked out a complex account of how objects produce sense impressions and explained error by positing the disruption of causal effluences in transit.

Saint Thomas Aquinas (1224-1274) held the view (mentioned above) that intellectual knowledge is derived by way of abstraction (concept formation) from sense data. Aquinas even argued that the existence of God could be proved by reasoning from sense data. This, he posited, could be accomplished via his version of the Aristotelian active intellect which he understood as the faculty of abstracting universal meanings from particular empirical data.

Francis Bacon (1561-1626) gave impetus to the development of modern inductive science. Of the earlier philosophers, he particularly criticized Aristotle. British empiricists took their cue from Bacon who hailed the primacy of experience, particularly over nature. Bacon relates the following story which demonstrates the difference between rationalism and empiricism:

Francis Bacon (1605): In the year of our Lord 1432, there arose a grievous quarrel among the brethren over the number of teeth in the mouth of a horse. For 13 days the disputation raged without ceasing. All the ancient books and chronicles were fetched out, and wonderful and ponderous erudition, such as was never before heard of in this region, was made manifest. At the beginning of the 14th day, a youthful friar of goodly bearing asked his learned superiors for permission to add a word, and straightaway, to the wonderment of the disputants, whose deep wisdom he sore vexed, he beseeched them to unbend in a manner coarse and unheard-of, and to look in the open mouth of a horse and find answer to their questionings. At this, their dignity being grievously hurt, they waxed exceedingly wroth; and joining in a mighty uproar, they flew upon him and smote him hip and thigh, and cast him out forthwith. For, said they, surely Satan hath tempted this bold neophyte to declare unholy and unheard-of ways of finding truth contrary to all the teaching of the fathers. After many days more of grievous strife the dove of peace sat on the assembly and they as one man, declaring the problem to be an everlasting mystery because of a grievous dearth of historical and theological evidence thereof, so ordered the same writ down. (Excerpted from Munn, (1951). *Introduction to Psychology*. Boston: Houghton-Mifflin.)

The British Empiricists

John Locke (1632-1704), the first and founder of the British empiricists, was an empiricist in roughly the same sense that Aquinas was. His main target for attack was the doctrine of innate ideas- the doctrine that there may be ideas with which we are born or, at any rate, which we do not have to derive from sense experience.

Let us suppose the mind to be, as we say, white paper, void of all characters, without any ideas; how comes it by that vast store, which the busy and boundless fancy of man has painted on it with an almost endless variety? Whence has it all the materials of reason and knowledge? To this I answer, in one word, from experience: in that all our knowledge is founded.

George Berkeley (1685-1753) was the second of the British empiricists. One of his aims was to rid Locke’s philosophy of those elements which were inconsistent with empiricism. The *esse* of sensible things is *percipi*-- they consist in being perceived and they have no existence without the mind (Idealism). Berkeley held that even subjects like geometry had to be limited in scope in order to rule out nonempirical objects of knowledge. Thus, Berkeley maintained that there is a least perceptible size; hence, there can be no ideas of infinitesimals or points. Berkeley asserted that knowledge is entirely dependent on sensations for all its materials other than the notions we have of God and ourselves. The certainty of our sensations is due to the fact that there can be no question whether they actually represent a reality behind them, and this is the basis of Berkeley’s claim to deal with skepticism.

David Hume (1711-1776) was a Scottish empiricist whose work in *Treatise of Human Nature* reveals the philosophical influence of John Locke and George Berkeley. Hume tried to improve on the work of his predecessors with attempts at greater precision. He distinguished first between impressions and ideas, the former being the contents of the mind in perception, the latter those in imagination, etc. He further subdivided ideas into those of sense and those of reflection, and again, into those which are simple and those which are complex. A cardinal point of his empiricism was that every simple idea is a copy of a corresponding impression. As an empiricist, Hume attempted to show how human knowledge arises from sense experience. His method led him to conclusions that were skeptical of many established beliefs. Perhaps his most famous discussions concern the idea of causality. Hume argued that belief in a necessary connection between cause and effect is based on habit and custom rather than reason or observation. His ideas have influenced Logical Positivism in the philosophy of science.

John Stuart Mill (1806-1873) left a permanent imprint on philosophy through his restatements of the principles underlying empiricism and utilitarianism. He followed directly in the tradition of Hume. Mill's account of our knowledge of the external world was in part phenomenalist in character; it maintained that things are merely permanent possibilities of sensation. Mill was more radical than Hume. He was so impressed by the possibilities of the use of induction that he found inductive inference in places where we should not ordinarily expect to find it. Mill claimed that mathematical truths were merely very highly confirmed generalizations from experience; mathematical inference, generally conceived as deductive in nature, he set down as founded on induction. This is perhaps the most extreme version of empiricism known, and it has not many followers.

Empiricism and the American Philosophers

Ralph Waldo Emerson (1803-1882) opposed the skepticism of Locke and the empiricists and is generally considered the leading exponent of American Transcendentalism. The general philosophical concept of transcendence, or belief in a higher reality not validated by sense experience or pure reason, was developed in ancient times by Parmenides and Plato. Emerson helped to start the *Transcendental Club* in 1836 and published *Nature* (1836), a book showing the organicism of all life and the function of nature as a visible manifestation of invisible spiritual truths. Emerson's transcendentalism is closely associated with the Idealism of Kant and is also a close approximation of European Romanticism. Emerson credits Kant with "showing that there was a very important class of ideas or imperative forms, which did not come by experience, but through which experience was acquired; that these were intuitions of the mind itself; and he denominated them Transcendental forms."

Charles Sanders Peirce (1839-1914) is perhaps best known not for his empiricism but as the founder of the pragmatic movement (Pragmatism) in American philosophy. He met William James at Harvard who later developed and popularized pragmatism. As regards empiricism, Peirce notes four methods for "fixing belief," belief being the goal of inquiry, and espouses the scientific or experimental method as the only truly successful method of fixing belief; it leads everyone who employs it ultimately to the same conclusion. This method presupposes that (1) the objects of knowledge are real things, (2) the characters (properties) of real things do not depend on our perceptions of them, and (3) everyone who has sufficient experience of real things will agree on the truth about them. According to Peirce's doctrine of *fallibilism*, the conclusions of science are always tentative. The rationality of the scientific method does not depend on the certainty of its conclusions, but on its *self-corrective* character: by continued application of the method science can detect and correct its own mistakes, and thus eventually lead to the discovery of truth.

William James (1842-1910) along with Peirce (see above) was one of the founders and leading proponents of Pragmatism. James considered pragmatism to be both a method for analyzing philosophic problems and a theory of truth. He also saw it as an extension of the empiricist attitude in that it turned away from abstract theory and fixed or absolute principles and toward concrete facts, actions, and relative principles. In a letter to Francois Pillon in 1904, James writes: *"My philosophy is what I call a radical empiricism, a pluralism, a 'tychism,' which represents order as being gradually won and always in the making."* He claims that there is only one "stuff" of which everything in the world is made and that "stuff" is "pure experience." Now this pure experience is not a single entity, but rather a collective name for all sensible natures. It is a name for all the "thats" which anywhere appear. *"To be radical, an empiricism must neither admit into its constructions any element that is not directly experienced, nor exclude from them any element that is directly experienced."* James takes it as a given that relations between things are equivalently experienced as the things themselves. James' radical empiricism finds connections between experiences in experience itself. There are the intellectual connections where experiences know or believe or remember other experiences. And then there are the non-intellectual connections such as cause and effect or the tendencies of one experience to follow another (ie. fire and smoke).

John Dewey (1859-1952) carried on the leadership of the pragmatist movement after James death. His version of pragmatism was called Instrumentalism. The key concept in Dewey's philosophy is experience. He thought of experience as a single, dynamic, unified whole in which everything is ultimately interrelated. At the highest level of generality one might call Dewey's philosophy a kind of *naturalistic empiricism*. Dewey thought of himself as part of a general movement that was developing a new empiricism based on a new concept of experience, one that combined the strong naturalistic bias of the Greek philosophers with a sensitive appreciation for experimental method as practiced by the sciences. His concept of experience had its origin in his Hegelian background, but Dewey divested it of most of its speculative excesses. He clearly conceived of himself as an empiricist but was careful to distinguish his notion of experience both from that of the idealist tradition and from the empiricism of the classical British variety. The idealists had so stressed the cognitive dimension of experience that they overlooked the non-cognitive, whereas he saw the British variety as inappropriately atomistic and subjectivist. In contrast to these Dewey fashioned a notion of experience wherein action, enjoyment, and what he called "undergoing" were integrated and equally fundamental.

Willard Van Orman Quine (June 25, 1908-). The empiricism of Quine is perhaps the most difficult to get a handle on. I suppose one might write Mr. Quine at Harvard and consult him on the matter. This might be the best method for ascertaining exactly what Quine means by empiricism (since we have seen that throughout the various philosophers there have been many empiricisms). In his *Two Dogmas of Empiricism* Quine deals with the "ill-founded" dogmas of *analytic* and *subjective* truths. I don't intend to give an analysis of that essay here, but simply wish to allude to Quine's empiricist attitude therein. Directly from the text of the essay: *"The totality of our so-called knowledge or beliefs, from the most causal matters of geography and history to the profoundest laws of atomic physics or even pure mathematics and logic, is a man-made fabric, which impinges on experience only along the edges. Or, to change the figure, total science is like a field of force whose boundary conditions are experience."* Quine maintains that any conflict with experience at the "edges" will alter conditions at the interior. And later on Quine states explicitly that *"As an empiricist I continue to think of the conceptual scheme of science as a tool, ultimately, for predicting future experience in the light of past experience."* Quine carries on with the idea stated above regarding the totality of science or of our beliefs as an interdependent and interconnected "web" in his *The Web of Belief,* co-authored by J.S. Ullian. Throughout this and others of his works, Quine's empiricist attitude may be encountered, however, I found the majority of his works deal with the regimentation of ordinary language (language and meaning) as opposed to a strict ontologic attitude. Quine calls theories regarding ontology "ontic theories." He sees that the integration of established theories may lead to any one of a number of equally satisfactory accounts of the world, each with its "ontic theory," and, according to Quine, it makes no sense to ask which one is true. Quine thus takes a conventionalist view regarding theses of ontology.

Empiricism is the theory that all knowledge stems from sense experience and internal mental experience- such as emotions and self-reflection. The empiricist draws his rules of practice not from theory but from close observation and experiment, emphasizing inductive rather than deductive processes of thought. For empiricists, facts precede theories and it is possible for one to be an impartial, objective observer of "facts." Empiricists claim that no one could have knowledge of the world unless he had experiences and could reason, but this does not mean that either experience or reason by themselves could provide a kind of absolute certainty about the world- but then, what can?

Teddy Ward

Jean Piaget (1896-1980) was a biologist who originally studied molluscs (publishing twenty scientific papers on them by the time he was 21) but moved into the study of the development of children's understanding, through observing them and talking and listening to them while they worked on exercises he set.

> "Piaget's work on children's intellectual development owed much to his early studies of water snails"
>
> (Satterly, 1987:622)

His view of how children's minds work and develop has been enormously influential, particularly in educational theory. His particular insight was the role of maturation (simply growing up) in children's increasing capacity to understand their world: they cannot undertake certain tasks until they are psychologically mature enough to do so. His research has spawned a great deal more, much of which has undermined the detail of his own, but like many other original investigators, his importance comes from his overall vision.

He proposed that children's thinking does not develop entirely smoothly: instead, there are certain points at which it "takes off" and moves into completely new areas and capabilities. He saw these transitions as taking place at about 18 months, 7 years and 11 or 12 years. This has been taken to mean that before these ages children are not capable (no matter how bright) of understanding things in certain ways, and has been used as the basis for scheduling the school curriculum. Whether or not *should* be the case is a different matter.

More

Piaget's Key Ideas

Adaptation	What it says: adapting to the world through assimilation and accommodation
Assimilation	The process by which a person takes material into their mind from the environment, which may mean changing the evidence of their senses to make it fit.
Accommodatio	The difference made to one's mind or concepts by the process of

n	assimilation. Note that assimilation and accommodation go together: you can't have one without the other.
Classification	The ability to group objects together on the basis of common features.
Class Inclusion	The understanding, more advanced than simple classification, that some classes or sets of objects are also sub-sets of a larger class. (E.g. there is a class of objects called dogs. There is also a class called animals. But all dogs are also animals, so the class of animals includes that of dogs)
Conservation	The realisation that objects or sets of objects stay the same even when they are changed about or made to look different.
Decentration	The ability to move away from one system of classification to another one as appropriate.
Egocentrism	The belief that you are the centre of the universe and everything revolves around you: the corresponding inability to see the world as someone else does and adapt to it. Not moral "selfishness", just an early stage of psychological development.
Operation	The process of working something out in your head. Young children (in the sensorimotor and pre-operational stages) have to act, and try things out in the real world, to work things out (like count on fingers): older children and adults can do more in their heads.
Schema (or scheme)	The representation in the mind of a set of perceptions, ideas, and/or actions, which go together.
Stage	A period in a child's development in which he or she is capable of understanding some things but not others

Stages of Cognitive Development

Stage	Characterised by
Sensori-motor (Birth-2 yrs)	Differentiates self from objects Recognises self as agent of action and begins to act intentionally: e.g. pulls a string to set mobile in motion or shakes a rattle to make a noise Achieves object permanence: realises that things continue to exist even when no longer present to the sense (pace Bishop Berkeley)

Pre-operational (2-7 years)	Learns to use language and to represent objects by images and words Thinking is still egocentric: has difficulty taking the viewpoint of others Classifies objects by a single feature: e.g. groups together all the red blocks regardless of shape or all the square blocks regardless of colour
Concrete operational (7-11 years)	Can think logically about objects and events Achieves conservation of number (age 6), mass (age 7), and weight (age 9) Classifies objects according to several features and can order them in series along a single dimension such as size.
Formal operational (11 years and up)	Can think logically about abstract propositions and test hypotheses systemtically Becomes concerned with the hypothetical, the future, and ideological problems

The accumulating evidence is that this scheme is too rigid: many children manage concrete operations earlier than he thought, and some people never attain formal operations (or at least are not called upon to use them).

Piaget's approach is central to the school of cognitive theory known as "cognitive constructivism": other scholars, known as "social constructivists", such as Vygotsky and Bruner, have laid more emphasis on the part played by language and other people in enabling children to learn.

See here for Howard Gardner's re-evaluation of Piaget: still a giant, but wrong in practically every detail.

And the combination of neuroscience and evolutionary psychology is beginning to suggest that the overall developmental model is based on dubious premises. (It's too early to give authoritative references for this angle.)

Jean Piaget's stages of cognitive development describe the intellectual development of children from infancy to early adulthood. Piaget believed that children are not less intelligent than adults, they simply think differently. He also proposed a number of concepts to explain how children process information.

- Key Concepts of Piaget's Theory
  Important concepts in Piaget's stages of cognitive development include assimilation, accommodation, and equilibration. Learn more about these concepts as well as the background of Piaget's theory.

- [The Sensorimotor Stage]
  The sensorimotor stage can be divided into six separate substages that are characterized by the development of a new skill.
- [The Preoperational Stage]
  This stage is characterized by an increase in playing and pretending. Characteristics of this stage include egocentrism and difficulty understanding conservation.
- [The Concrete Operational Stage]
  During this stage, children begin thinking logically about concrete events, but have difficulty understanding abstract or hypothetical concepts.
- [The Formal Operational Stage]
  During this stage of cognitive development, skills such as logical thought, deductive reasoning, and systematic planning begin to emerge.
- [Support and Criticism of Piaget's Theory]
  While Piaget's stage theory of cognitive development has been influential in psychology, there have been a number of criticims of his work. Learn more about support and criticism of Piaget's theory.

- Piaget's stage theory describes the cognitive development of children. Cognitive development involves changes in co
  Piaget's view, early cognitive development involves processes based upon actions and later progresses into changes
- **Key Concepts**
- **Schemas** - A schema describes both the mental and physical actions involved in understanding and knowing. Schem
  help us to interpret and understand the world. In Piaget's view, a schema includes both a category of knowledge an
  knowledge. As experiences happen, this new information is used to modify, add to, or change previously existing sc
  a schema about a type of animal, such as a dog. If the child's sole experience has been with small dogs, a child mig
  furry, and have four legs. Suppose then that the child encounters a very large dog. The child will take in this new in
  existing schema to include this new information.
- **Assimilation** - The process of taking in new information into our previously existing schema's is known as assimilat
  subjective, because we tend to modify experience or information somewhat to fit in with our preexisting beliefs. In t
  labeling it "dog" is an example of assimilating the animal into the child's dog schema.
- **Accommodation** - Another part of adaptation involves changing or altering our existing schemas in light of new inf
  accommodation. Accommodation involves altering existing schemas, or ideas, as a result of new information or new
  be developed during this process.
- **Equilibration** - Piaget believed that all children try to strike a balance between assimilation and accommodation, w
  Piaget called equilibration. As children progress through the stages of cognitive development, it is important to main
  previous knowledge (assimilation) and changing behavior to account for new knowledge (accommodation). Equilibra
  able to move from one stage of thought into the next.
- **Characteristics of the Sensorimotor Stage:**
- The first stage of Piaget's theory lasts from birth to approximately age two and is centered on the infant trying to m
  sensorimotor stage, an infant's knowledge of the world is limited to their sensory perceptions and motor activities. B
  responses caused by sensory stimuli. Children utilize skills and abilities they were born with, such as looking, suckin
  more about the environment.
- **Substages of the Sensorimotor Stage:**
- The sensorimotor stage can be divided into six separate substages that are characterized by the development of a n
- **Reflexes (0-1 month):**
- During this substage, the child understands the environment purely through inborn reflexes such as sucking and loo
- **Primary Circular Reactions (1-4 months):**
- This substage involves coordinating sensation and new schemas. For example, a child may such his or her thumb by
  repeat the action. These actions are repeated because the infant finds them pleasurable.
- **Secondary Circular Reactions (4-8 months):**
- During this substage, the child becomes more focused on the world and begins to intentionally repeat an action in o
  environment. For example, a child will purposefully pick up a toy in order to put it in his or her mouth.
- **Coordination of Reactions (8-12 months):**
- During this substage, the child starts to show clearly intentional actions. The child may also combine schemas in ord
  begin exploring the environment around them and will often imitate the observed behavior of others. The understan
  time and children begin to recognize certain objects as having specific qualities. For example, a child might realize t
  shaken.
- **Tertiary Circular Reactions (12-18 months):**

- Children begin a period of trial-and-error experimentation during the fifth substage. For example, a child may try ou of getting attention from a caregiver.
- **Early Representational Thought (18-24 months):**
- Children begin to develop symbols to represent events or objects in the world in the final sensorimotor substage. Du towards understanding the world through mental operations rather than purely through actions.

Social Development Theory (Vygotsky)

Summary: Social Development Theory argues that social interaction precedes development; consciousness and cognition are the end product of socialization and social behavior.

Originator: Lev Vygotsky (1896-1934).

Key terms: Zone of Proximal Development (ZPD), More Knowledgeable Other (MKO)

Vygotsky's Social Development Theory

Vygotsky's Social Development Theory is the work of Russian psychologist Lev Vygotsky (1896-1934), who lived during Russian Revolution. Vygotsky's work was largely unkown to the West until it was published in 1962.

Vygotsky's theory is one of the foundations of constructivism. It asserts three major themes:

Major themes:

1. Social interaction plays a fundamental role in the process of cognitive development. In contrast to Jean Piaget's understanding of child development (in which development necessarily precedes learning), Vygotsky felt social learning precedes development. He states: "Every function in the child's cultural development appears twice: first, on the social level, and later, on the individual level; first, between people (interpsychological) and then inside the child (intrapsychological)." (Vygotsky, 1978).
2. The More Knowledgeable Other (MKO). The MKO refers to anyone who has a better understanding or a higher ability level than the learner, with respect to a particular task, process, or concept. The MKO is normally thought of as being a teacher, coach, or older adult, but the MKO could also be peers, a younger person, or even computers.
3. The Zone of Proximal Development (ZPD). The ZPD is the distance between a student's ability to perform a task under adult guidance and/or with peer collaboration and the student's ability solving the problem independently. According to Vygotsky, learning occurred in this zone.

Vygotsky focused on the connections between people and the sociocultural context in which they act and interact in shared experiences (Crawford, 1996). According to Vygotsky, humans use tools that develop from a culture, such as speech and writing, to mediate their social environments. Initially children develop these tools to serve solely as social functions, ways to communicate needs. Vygotsky believed that the internalization of these tools led to higher thinking skills.

Applications of the Vygotsky's Social Development Theory

Many schools have traditionally held a transmissionist or instructionist model in which a teacher or lecturer 'transmits' information to students. In contrast, Vygotsky's theory promotes learning contexts in which students play an active role in learning. Roles of the teacher and student are therefore shifted, as a teacher should collaborate with his or her students in order to help facilitate meaning construction in students. Learning therefore becomes a reciprocal experience for the students and teacher.

A pioneering psychologist, Vygotsky was also a highly prolific author: his major works span 6 volumes, written over roughly 10 years, from his Psychology of Art (1925) to Thought and Language [or Thinking and Speech] (1934). Vygotsky's interests in the fields of developmental psychology, child development, and education were extremely diverse. The philosophical framework he provided includes not only insightful interpretations about the cognitive role of tools of mediation, but also the re-interpretation of well-known concepts in psychology such as the notion of internalization of knowledge. Vygotsky introduced the notion of zone of proximal development, an innovative metaphor capable of describing not the actual, but the potential of human cognitive development. His work covered such diverse topics as the origin and the psychology of art, development of higher mental functions, philosophy of science and methodology of psychological research, the relation between learning and human development, concept formation, interrelation between language and thought development, play as a psychological phenomenon, the study of learning disabilities, and abnormal human development (aka *defectology*).

[edit] Cultural mediation and internalization

Vygotsky investigated child development and how this was guided by the role of culture and interpersonal communication. Vygotsky observed how higher mental functions developed historically within particular cultural groups, as well as individually through social interactions with significant people in a child's life, particularly parents, but also other adults. Through these interactions, a child came to learn the habits of mind of her/his culture, including speech patterns, written language, and other symbolic knowledge through which the child derives meaning and which affected a child's construction of her/his knowledge. This key premise of Vygotskian psychology is often referred to as *cultural mediation*. The specific knowledge gained by children through these interactions also represented the shared knowledge of a culture. This process is known as internalization.[3]

Internalization can be understood in one respect as "knowing how". For example, riding a bicycle or pouring a cup of milk are tools of the society and initially outside and beyond the child. The mastery of these skills occurs through the activity of the child within society. A further aspect of internalization is appropriation, in which the child takes a tool and makes it his own, perhaps using it in a way unique to himself. Internalizing the use of a pencil allows the child to use it very much for his own ends rather than draw exactly what others in society have drawn previously.[3]

Guided participation, which takes place when creative thinkers interact with a knowledgeable person, is practiced around the world. Cultures may differ, though, in the goals of development. For example, Mayan mothers in Guatemala help their daughters learn to weave through guided participation.[3]

[edit] Psychology of play

Less known is Vygotsky's research on play, or children's games, as a psychological phenomenon and its role in the child's development. Through play the child develops abstract meaning separate from the objects in the world, which is a critical feature in the development of higher mental functions.[4]

The famous example Vygotsky gives is of a child who wants to ride a horse but cannot. If the child were under three, he would perhaps cry and be angry, but around the age of three the child's relationship with the world changes: "Henceforth play is such that the explanation for it must always be that it is the imaginary, illusory realization of unrealizable desires. Imagination is a new formation that is not present in the consciousness of the very raw young child, is totally absent in animals, and represents a specifically human form of conscious activity. Like all functions of consciousness, it originally arises from action." (Vygotsky, 1978)

The child wishes to ride a horse but cannot, so he picks up a stick and stands astride of it, thus pretending he is riding a horse. The stick is a *pivot*. "Action according to rules begins to be determined by ideas, not by objects.... It is terribly difficult for a child to sever thought (the meaning of a word) from object. Play is a transitional stage in this direction. At that critical moment when a stick – i.e., an object – becomes a pivot for severing the meaning of horse from a real horse, one of the basic psychological structures determining the child's relationship to reality is radically altered".

As children get older, their reliance on pivots such as sticks, dolls and other toys diminishes. They have *internalized* these pivots as imagination and abstract concepts through which they can understand the world. "The old adage that children's play is imagination in action can be reversed: we can say that imagination in adolescents and schoolchildren is play without action" (Vygotsky, 1978).

Another aspect of play that Vygotsky referred to was the development of social rules that develop, for example, when children play house and adopt the roles of different family members. Vygotsky cites an example of two sisters playing at being sisters. The rules of behavior between them that go unnoticed in daily life are consciously acquired through play. As well as social rules, the child acquires what we now refer to as self-regulation. For example, when a child stands at the starting line of a running race, she may well desire to run immediately so as to reach the finish line first, but her knowledge of the social rules surrounding the game and her desire to enjoy the game enable her to regulate her initial impulse and wait for the start signal.

[edit] *Thought and Language*

Perhaps Vygotsky's most important contribution concerns the inter-relationship of language development and thought. This concept, explored in Vygotsky's book *Thought and Language*, (alternative translation: *Thinking and Speaking*) establishes the explicit and profound connection between speech (both silent inner speech and oral language), and the development of mental concepts and cognitive awareness. It should be noted that Vygotsky described inner speech as being qualitatively different from normal (external) speech. Although Vygotsky believed inner speech developed from external speech via a gradual process of internalization, with younger children only really able to "think out loud," he claimed that in its mature form inner speech would be unintelligible to anyone except the thinker, and would not resemble spoken language as we know it (in particular, being greatly compressed). Hence, thought itself develops socially.[3]

An infant learns the meaning of signs through interaction with its main care-givers, e.g., pointing, cries, and gurgles can express what is wanted. How verbal sounds can be used to conduct social interaction is learned through this activity, and the child begins to utilize, build, and develop this faculty, e.g., using names for objects, etc.[3]

Language starts as a tool external to the child used for social interaction. The child guides personal behavior by using this tool in a kind of self-talk or "thinking out loud." Initially, self-talk is very much a tool of social interaction and it tapers to negligible levels when the child is alone or with deaf children. Gradually self-talk is used more as a tool for self-directed and self-regulating behavior. Then, because speaking has been appropriated and internalized, self-talk is no longer present around the time the child starts school. Self-talk "develops along a rising not a declining, curve; it goes through an evolution, not an involution. In the end, it becomes inner speech" (Vygotsky, 1987, pg 57). Inner speech develops through its differentiation from social speech.[3]

Speaking has thus developed along two lines, the line of social communication and the line of inner speech, by which the child mediates and regulates their activity through their thoughts which in turn are mediated by the semiotics (the meaningful signs) of inner speech. This is not to say that thinking cannot take place without language, but rather that it is mediated by it and thus develops to a much higher level of sophistication. Just as the birthday cake as a sign provides much deeper meaning than its physical properties allow, inner speech as signs provides much deeper meaning than the lower psychological functions would otherwise allow.[3]

Inner speech is not comparable in form to external speech. External speech is the process of turning thought into words. Inner speech is the opposite; it is the conversion of speech into inward thought. Inner speech for example contains predicates only. Subjects are superfluous. Words too are used much more economically. One word in inner speech may be so replete with sense to the individual that it would take many words to express it in external speech.[3]

[edit] Zone of proximal development

"Zone of proximal development" (ZPD) is Vygotsky's term for the range of tasks that a child can complete independently and those completed with the guidance and assistance of adults or more-skilled children. The lower limit of ZPD is the level of skill reached by the child working independently. The upper limit is the level of additional responsibility the child can accept with the assistance of an able instructor. The ZPD captures the child's cognitive skills that are in the process of maturing and can be accomplished only with the assistance of a more-skilled person. Scaffolding is a concept closely related to the idea of ZPD. Scaffolding is changing the level of support. Over the course of a teaching session, a more-skilled person adjusts the amount of guidance to fit the child's current performance. Dialogue is an important tool of this process in the zone of proximal development. In a dialogue unsystematic, disorganized, and spontaneous concepts of a child are met with the more systematic, logical and rational concepts of the skilled helper.[3]

Moral development is a topic of interest in both psychology and education. Psychologist Lawrence Kohlberg modified and expanded upon Jean Piaget's work to form a theory that explained the development of moral reasoning. Piaget described a two-stage process of moral development, while Kohlberg theory of moral development outlined six stages within three different levels. Kohlberg extended Piaget's theory, proposing that moral development is a continual process that occurs throughout the lifespan.

"The Heinz Dilemma"

Kohlberg based his theory upon research and interviews with groups of young children. A series of moral dilemmas were presented to children, who were then interviewed to determine the reasoning behind their judgments of each scenario. The following is one example of the dilemmas Kohlberg presented.

"**Heinz Steals the Drug**
In Europe, a woman was near death from a special kind of cancer. There was one drug that the doctors thought might save her. It was a form of radium that a druggist in the same town had recently discovered. The drug was expensive to make, but the druggist was charging ten times what the drug cost him to make. He paid $200 for the radium and charged $2,000 for a small dose of the drug.

The sick woman's husband, Heinz, went to everyone he knew to borrow the money, but he could only get together about $ 1,000 which is half of what it cost. He told the druggist that his wife was dying and asked him to sell it cheaper or let him pay later. But the druggist said: "No, I discovered the drug and I'm going to make money from it." So Heinz got desperate and broke into the man's store to steal the drug-for his wife. Should the husband have done that? (Kohlberg, 1963)."

Kohlberg was not interested so much in the answer to the question of whether Heinz was wrong or right, but in the *reasoning* for the participants decision. The responses were then classified into various stages of reasoning in his theory of moral development.

Level 1. Preconventional Morality

- **Stage 1 - Obedience and Punishment**
  The earliest stage of moral development is especially common in young children, but adults are also capable of expressing this type of reasoning. At this stage, children see rules as fixed and absolute. Obeying the rules is important because it is a means to avoid punishment.

- **Stage 2 - Individualism and Exchange**
  At this stage of moral development, children account for individual points of view and judge actions based on how they serve individual needs. In the Heinz dilemma, children argued that the best course of action was the choice that best-served Heinz’s needs. Reciprocity is possible, but only if it serves one's own interests.

Level 2. Conventional Morality

- **Stage 3 - Interpersonal Relationships**
  Often referred to as the "good boy-good girl" orientation, this stage of moral development is focused on living up to social expectations and roles. There is an emphasis on conformity, being "nice," and consideration of how choices influence relationships.

- **Stage 4 - Maintaining Social Order**
  At this stage of moral development, people begin to consider society as a whole when making judgments. The focus is on maintaining law and order by following the rules, doing one’s duty and respecting authority.

Level 3. Postconventional Morality

- **Stage 5 - Social Contract and Individual Rights**
  At this stage, people begin to account for the differing values, opinions and beliefs of other people. Rules of law are important for maintaining a society, but members of the society should agree upon these standards.

- **Stage 6 - Universal Principles**
  Kolhberg's final level of moral reasoning is based upon universal ethical principles and abstract reasoning. At this stage, people follow these internalized principles of justice, even if they conflict with laws and rules.

Criticisms of Kohlberg's Theory of Moral Development:

- Does moral reasoning necessarily lead to moral behavior? Kohlberg's theory is concerned with moral thinking, but there is a big difference between knowing what we *ought* to do versus our actual actions.

- Is justice the only aspect of moral reasoning we should consider? Critics have pointed out that Kohlberg's theory of moral development overemphasizes the concept as justice when making moral choices. Factors such as compassion, caring and other interpersonal feelings may play an important part in moral reasoning.

- Does Kohlberg's theory overemphasize Western philosophy? Individualistic cultures emphasize personal rights while collectivist cultures stress the importance of society and community. Eastern cultures may have different moral outlooks that Kohlberg's theory does not account for.
- **Kohlberg's stages of moral development** constitute an adaptation of a psychological theory originally conceived of by the Swiss psychologist Jean Piaget. Lawrence Kohlberg began work on this topic while a psychology postgraduate student at the University of Chicago,[1] and expanded and developed this theory throughout the course of his life.
- The theory holds that moral reasoning, the basis for ethical behavior, has six identifiable developmental stages, each more adequate at responding to moral dilemmas than its predecessor.[2] Kohlberg followed the development of moral judgment far beyond the ages studied earlier by Piaget,[3] who also claimed that logic and morality develop through constructive stages.[2] Expanding on Piaget's work, Kohlberg determined that the process of moral development was principally concerned with justice, and that it continued throughout the individual's lifetime,[4] a notion that spawned dialogue on the philosophical implications of such research.[5][6]
- Kohlberg relied for his studies on stories such as the Heinz dilemma, and was interested in how individuals would justify their actions if placed in similar moral dilemmas. He then analyzed the form of moral reasoning displayed, rather than its conclusion,[6] and classified it as belonging to one of six distinct stages.[7][8][9]
- There have been critiques of the theory from several perspectives. Arguments include that it emphasizes justice to the exclusion of other moral values, such as caring;[10] that there is such an overlap between stages that they should more properly be regarded as separate domains; or that evaluations of the reasons for moral choices are mostly *post hoc* rationalizations (by both decision makers and psychologists studying them) of essentially intuitive decisions.
- Nevertheless, an entirely new field within psychology was created as a direct result of Kohlberg's theory, and according to Haggbloom et al.'s study of the most eminent psychologists of the 20th century, Kohlberg was the 16th most frequently cited psychologist in introductory psychology textbooks throughout the century, as well as the 30th most eminent overall.[11]
- Kohlberg's scale is about how people justify behaviors and his stages are not a method of ranking how moral someone's behavior is. There should however be a correlation between how someone scores on the scale and how they behave and the general hypothesis is that

moral behaviour is more responsible, consistent and predictable from people at higher levels.[12]

Stages

Kohlberg's six stages can be more generally grouped into three levels of two stages each: pre-conventional, conventional and post-conventional.[7][8][9] Following Piaget's constructivist requirements for a stage model, as described in his theory of cognitive development, it is extremely rare to regress backward in stages—to lose the use of higher stage abilities.[13][14] Stages cannot be skipped; each provides a new and necessary perspective, more comprehensive and differentiated than its predecessors but integrated with them.[13][14]

Level 1 (Pre-Conventional)

1. Obedience and punishment orientation

(*How can I avoid punishment?*)

2. Self-interest orientation

(*What's in it for me?*)

Level 2 (Conventional)

3. Interpersonal accord and conformity

(*Social norms*)

(*The good boy/good girl attitude*)

4. Authority and social-order maintaining orientation

(*Law and order morality*)

Level 3 (Post-Conventional)

5. Social contract orientation

6. Universal ethical principles

(*Principled conscience*)

[edit] Pre-Conventional

The pre-conventional level of moral reasoning is especially common in children, although adults can also exhibit this level of reasoning. Reasoners at this level judge the morality of an action by its direct consequences. The pre-conventional level consists of the first and second stages of moral development, and is solely concerned with the self in an egocentric manner. A child with preconventional morality has not yet adopted or internalized society's conventions regarding what is right or wrong, but instead focuses largely on external consequences that certain actions may bring.[7][8][9]

In **Stage one** (obedience and punishment driven), individuals focus on the direct consequences of their actions on themselves. For example, an action is perceived as morally wrong because the perpetrator is punished. "The last time I did that I got spanked so I will not do it again." The worse the punishment for the act is, the more "bad" the act is perceived to be.[15] This can give rise to an inference that even innocent victims are guilty in proportion to their suffering. It is "egocentric", lacking recognition that others' points of view are different from one's own.[16] There is "deference to superior power or prestige".[16]

Stage two (self-interest driven) espouses the "what's in it for me" position, in which right behavior is defined by whatever is in the individual's best interest. Stage two reasoning shows a limited interest in the needs of others, but only to a point where it might further the individual's own interests. As a result, concern for others is not based on loyalty or intrinsic respect, but rather a "you scratch my back, and I'll scratch yours" mentality.[2] The lack of a societal perspective in the pre-conventional level is quite different from the social contract (stage five), as all actions have the purpose of serving the individual's own needs or interests. For the stage two theorist, the world's perspective is often seen as morally relative.

[edit] Conventional

The conventional level of moral reasoning is typical of adolescents and adults. Those who reason in a conventional way judge the morality of actions by comparing them to society's views and expectations. The conventional level consists of the third and fourth stages of moral development. Conventional morality is characterized by an acceptance of society's conventions concerning right and wrong. At this level an individual obeys rules and follows society's norms even when there are no consequences for obedience or disobedience. Adherence to rules and conventions is somewhat rigid, however, and a rule's appropriateness or fairness is seldom questioned.[7][8][9]

In **Stage three** (interpersonal accord and conformity driven), the self enters society by filling social roles. Individuals are receptive to approval or disapproval from others as it reflects society's accordance with the perceived role. They try to be a "good boy" or "good girl" to live up to these expectations,[2] having learned that there is inherent value in doing so. Stage three reasoning may judge the morality of an action by evaluating its consequences in terms of a person's relationships, which now begin to include things like respect, gratitude and the "golden rule". "I want to be liked and thought well of; apparently, not being naughty makes people like me." Desire to maintain rules and authority exists only to further support these social roles. The intentions of actions play a more significant role in reasoning at this stage; "they mean well ...".[2]

In **Stage four** (authority and social order obedience driven), it is important to obey laws, dictums and social conventions because of their importance in maintaining a functioning society. Moral reasoning in stage four is thus beyond the need for individual approval exhibited in stage three; society must learn to transcend individual needs. A central ideal or ideals often prescribe what is right and wrong, such as in the case of fundamentalism. If one person violates a law, perhaps everyone would—thus there is an obligation and a duty to uphold laws and rules. When someone does violate a law, it is morally wrong; culpability is thus a significant factor in this stage as it separates the bad domains from the good ones. Most active members of society remain at stage four, where morality is still predominantly dictated by an outside force.[2]

[edit] Post-Conventional

The post-conventional level, also known as the principled level, consists of stages five and six of moral development. There is a growing realization that individuals are separate entities from society, and that the individual's own perspective may take precedence over society's view; they may disobey rules inconsistent with their own principles. These people live by their own abstract principles about right and wrong—principles that typically include such basic human rights as life, liberty, and justice. Because of this level's "nature of self before others", the behavior of post-conventional individuals, especially those at stage six, can be confused with that of those at the pre-conventional level.

People who exhibit postconventional morality view rules as useful but changeable mechanisms—ideally rules can maintain the general social order and protect human rights. Rules are not absolute dictates that must be obeyed without question. Contemporary theorists often speculate that many people may never reach this level of abstract moral reasoning.[7][8][9]

In **Stage five** (social contract driven), the world is viewed as holding different opinions, rights and values. Such perspectives should be mutually respected as unique to each person or community. Laws are regarded as social contracts rather than rigid edicts. Those that do not promote the general welfare should be changed when necessary to meet "the greatest good for the greatest number of people".[8] This is achieved through majority decision, and inevitable compromise. Democratic government is ostensibly based on stage five reasoning.

In **Stage six** (universal ethical principles driven), moral reasoning is based on abstract reasoning using universal ethical principles. Laws are valid only insofar as they are grounded in justice, and a commitment to justice carries with it an obligation to disobey unjust laws. Rights are unnecessary, as social contracts are not essential for deontic moral action. Decisions are not reached hypothetically in a conditional way but rather categorically in an absolute way, as in the philosophy of Immanuel Kant.[17] This involves an individual imagining what they would do in another's shoes, if they believed what that other person imagines to be true.[18] The resulting consensus is the action taken. In this way action is never a means but always an end in itself; the individual acts because it is right, and not because it is instrumental, expected, legal, or previously agreed upon. Although Kohlberg insisted that stage six exists, he found it difficult to identify individuals who consistently operated at that level.[14]

[edit] Further stages

In Kohlberg's empirical studies of individuals throughout their life Kohlberg observed that some had apparently undergone moral stage regression. This could be resolved either by allowing for moral regression or by extending the theory. Kohlberg chose the latter, postulating the existence of sub-stages in which the emerging stage has not yet been fully integrated into the personality.[8] In particular Kohlberg noted a stage 4½ or 4+, a transition from stage four to stage five, that shared characteristics of both.[8] In this stage the individual is disaffected with the arbitrary nature of law and order reasoning; culpability is frequently turned from being defined by society to viewing society itself as culpable. This stage is often mistaken for the moral relativism of stage two, as the individual views those interests of society that conflict with their own as being relatively and morally wrong.[8] Kohlberg noted that this was often observed in students entering college.[8][14]

Kohlberg suggested that there may be a seventh stage—Transcendental Morality, or Morality of Cosmic Orientation—which linked religion with moral reasoning.[19] Kohlberg's difficulties in obtaining empirical evidence for even a sixth stage,[14] however, led him to emphasize the speculative nature of his seventh stage.[5]

[edit] Theoretical assumptions (philosophy)

The picture of human nature Kohlberg begins with is that humans are inherently communicative and capable of reason. They also possess a desire to understand others and the world around them. The stages of Kohlberg's model relate to the qualitative moral *reasonings* adopted by individuals, and so do not translate directly into praise or blame of any individual's actions or character. Arguing that his theory measures moral reasoning and not particular moral conclusions, Kohlberg insists that the *form and structure* of moral arguments is independent of the *content* of those arguments, a position he calls "formalism".[6][7]

Kohlberg's theory centers on the notion that justice is the essential characteristic of moral reasoning. Justice itself relies heavily upon the notion of sound reasoning based on principles. Despite being a justice-centered theory of morality, Kohlberg considered it to be compatible with plausible formulations of deontology[17] and eudaimonia.

Kohlberg's theory understands values as a critical component of the right. Whatever the right is, for Kohlberg, it must be universally valid across societies (a position known as "moral universalism"):[7] there can be no relativism. Moreover, morals are not natural features of the world; they are prescriptive. Nevertheless, moral judgments can be evaluated in logical terms of truth and falsity.

According to Kohlberg: someone progressing to a higher stage of moral reasoning cannot skip stages. For example, an individual cannot jump from being concerned mostly with peer judgments (stage three) to being a proponent of social contracts (stage five).[14] On encountering a moral dilemma and finding their current level of moral reasoning unsatisfactory, however, an individual will look to the next level. Realizing the limitations of the current stage of thinking is the driving force behind moral development, as each progressive stage is more adequate than the last.[14] The process is therefore considered to be constructive, as it is initiated by the conscious construction of the individual, and is not in any meaningful sense a component of the individual's innate dispositions, or a result of past inductions.

Bloom's Taxonomy is a classification of learning objectives within education. It refers to a classification of the different objectives that educators set for students (learning objectives). The taxonomy was first presented in 1956 through the publication *The Taxonomy of Educational Objectives, The Classification of Educational Goals, Handbook I: Cognitive Domain*, by Benjamin Bloom (editor), M. D. Englehart, E. J. Furst, W. H. Hill, and David Krathwohl. It is considered to be a foundational and essential element within the education community as evidenced in the 1981 survey *Significant writings that have influenced the curriculum: 1906-1981*, by H. G. Shane and the 1994 yearbook of the National Society for the Study of Education.

A great mythology has grown around the taxonomy, possibly due to many people learning about the taxonomy through second hand information. Bloom himself considered the Handbook, "one of the most widely cited yet least read books in American education".[1]

Domains

Key to understanding the taxonomy and its revisions, variations, and addenda over the years is an understanding that the original Handbook was intended only to focus on one of the three domains (as indicated in the domain specification in title), but there was expectation that additional material would be generated for the other domains (as indicated in the numbering of the handbook in the title). Bloom also considered the initial effort to be a starting point, as evidenced in a memorandum from 1971 in which he said, "Ideally each major field should have its own taxonomy in its own language - more detailed, closer to the special language and thinking of its experts, reflecting its own appropriate sub-divisions and levels of education, with possible new categories, combinations of categories and omitting categories as appropriate."[2]

Bloom's Taxonomy divides educational objectives into three "domains": Affective, Psychomotor, and Cognitive. Within the domains, learning at the higher levels is dependent on having attained prerequisite knowledge and skills at lower levels (Orlich, et al. 2004). A goal of Bloom's Taxonomy is to motivate educators to focus on all three domains, creating a more holistic form of education.

[edit] Affective

Skills in the **affective domain** describe the way people react emotionally and their ability to feel another living thing's pain or joy. Affective objectives typically target the awareness and growth in attitudes, emotion, and feelings.

There are five levels in the affective domain moving through the lowest order processes to the highest:

Receiving

> The lowest level; the student passively pays attention. Without this level no learning can occur.

Responding

> The student actively participates in the learning process, not only attends to a stimulus; the student also reacts in some way.

Valuing

> The student attaches a value to an object, phenomenon, or piece of information.

Organizing

> The student can put together different values, information, and ideas and accommodate them within his/her own schema; comparing, relating and elaborating on what has been learned.

Characterizing

> The student holds a particular value or belief that now exerts influence on his/her behaviour so that it becomes a characteristic.

[edit] Psychomotor

Skills in the **psychomotor domain** describe the ability to physically manipulate a tool or instrument like a hand or a hammer. Psychomotor objectives usually focus on change and/or development in behavior and/or skills.

Bloom and his colleagues never created subcategories for skills in the psychomotor domain, but since then other educators have created their own psychomotor taxonomies. [3] Simpson (1972) among other contributors, such as Harrow (1972) and Dave (1975) created a Psychomotor Taxonomy that helps to explain the evolution in the dextrerity on the physical movements, whether in normal people or high performance athletes. The proposed levels are:

1. **Perception**: *The ability to use sensory cues to guide motor activity*. This ranges from sensory stimulation, through cue selection, to translation. Examples: Detects non-verbal communication cues. Estimate where a ball will land after it is thrown and then moving to the correct location to catch the ball. Adjusts heat of stove to correct temperature by smell and taste of food. Adjusts the height of the forks on a forklift by comparing where the forks are in relation to the pallet. Key Words: chooses, describes, detects, differentiates, distinguishes, identifies, isolates, relates, selects.

2. **Set**: Readiness to act. *It includes mental, physical, and emotional sets. These three sets are dispositions that predetermine a person's response to different situations* (sometimes called mindsets). Examples: Knows and acts upon a sequence of steps in a manufacturing process. Recognize one's abilities and limitations. Shows desire to learn a new process (motivation). NOTE: This subdivision of Psychomotor is closely related with the “Responding to phenomena” subdivision of the Affective domain. Key Words: begins, displays, explains, moves, proceeds, reacts, shows, states, volunteers.

3. **Guided Response**: *The early stages in learning a complex skill that includes imitation and trial and error. Adequacy of performance is achieved by practicing*. Examples: Performs a mathematical equation as demonstrated. Follows instructions to build a model. Responds hand-signals of instructor while learning to operate a forklift. Key Words: copies, traces, follows, react, reproduce, responds

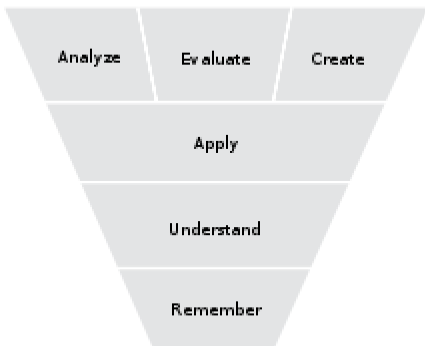
4. **Mechanism**: This is the intermediate stage in learning a complex skill. *Learned responses have become habitual and the movements can be performed with some confidence and proficiency*. Examples: Use a personal computer. Repair a leaking faucet. Drive a car. Key Words: assembles, calibrates, constructs, dismantles, displays, fastens, fixes, grinds, heats, manipulates, measures, mends, mixes, organizes, sketches.

5. **Complex Overt Response**: *The skillful performance of motor acts that involve complex movement patterns*. Proficiency is indicated by a quick, accurate, and highly coordinated performance, requiring a minimum of energy. This category includes performing without hesitation, and automatic performance. For example, players are often utter sounds of satisfaction or expletives as soon as they hit a tennis ball or throw a football, because they can tell by the feel of the act what the result will produce. Examples: Maneuvers a car into a tight parallel parking spot. Operates a computer quickly and accurately. Displays competence while playing the piano. Key Words: assembles, builds, calibrates, constructs, dismantles, displays, fastens, fixes, grinds, heats, manipulates, measures, mends, mixes, organizes, sketches. NOTE: The Key Words are the same as Mechanism, but will have adverbs or adjectives that indicate that the performance is quicker, better, more accurate, etc.

6. **Adaptation**: *Skills are well developed and the individual can modify movement patterns to fit special requirements*. Examples: Responds effectively to unexpected experiences. Modifies instruction to meet the needs of the learners. Perform a task with a machine that it was not originally intended to do (machine is not damaged and there is no danger in performing the new task). Key Words: adapts, alters, changes, rearranges, reorganizes, revises, varies.

7. **Origination**: *Creating new movement patterns to fit a particular situation or specific problem*. Learning outcomes emphasize creativity based upon highly developed skills. Examples: Constructs a new theory. Develops a new and comprehensive training programming. Creates a new gymnastic routine. Key Words: arranges, builds, combines, composes, constructs, creates, designs, initiate, makes, originates.

[edit] Cognitive



Categories in the cognitive domain of Bloom's Taxonomy (Anderson & Krathwohl, 2001)

Skills in the **cognitive domain** revolve around knowledge, comprehension, and critical thinking of a particular topic. Traditional education tends to emphasize the skills in this domain, particularly the lower-order objectives.

There are six levels in the taxonomy, moving through the lowest order processes to the highest:

Knowledge

> Exhibit memory of previously-learned materials by recalling facts, terms, basic concepts and answers
>
> - Knowledge of specifics - terminology, specific facts
> - Knowledge of ways and means of dealing with specifics - conventions, trends and sequences, classifications and categories, criteria, methodology
> - Knowledge of the universals and abstractions in a field - principles and generalizations, theories and structures

Questions like: What are the health benefits of eating apples?

Comprehension

> Demonstrative understanding of facts and ideas by organizing, comparing, translating, interpreting, giving descriptions, and stating main ideas

- Translation
- Interpretation
- Extrapolation

Questions like: Compare the health benefits of eating apples vs. oranges.

Application

> Using new knowledge. Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way

Questions like: Which kinds of apples are best for baking a pie, and why?

Analysis

> Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations

- Analysis of elements
- Analysis of relationships
- Analysis of organizational principles

Questions like: List four ways of serving foods made with apples and explain which ones have the highest health benefits. Provide references to support your statements.

Synthesis

> Compile information together in a different way by combining elements in a new pattern or proposing alternative solutions

- Production of a unique communication
- Production of a plan, or proposed set of operations
- Derivation of a set of abstract relations

Questions like: Convert an "unhealthy" recipe for apple pie to a "healthy" recipe by replacing your choice of ingredients. Explain the health benefits of using the ingredients you chose vs. the original ones.

Evaluation

> Present and defend opinions by making judgments about information, validity of ideas or quality of work based on a set of criteria

- Judgments in terms of internal evidence
- Judgments in terms of external criteria

Questions like: Do you feel that serving apple pie for an after school snack for children is healthy? Why or why not?

Some critiques of Bloom's Taxonomy's (cognitive domain) admit the existence of these six categories, but question the existence of a sequential, hierarchical link.[4] Also the revised edition of Bloom's taxonomy has moved Synthesis in higher order than Evaluation. Some consider the three lowest levels as hierarchically ordered, but the three higher levels as parallel.[5] Others say that it is sometimes better to move to Application before introducing concepts[citation needed]. This thinking would seem to relate to the method of problem-based learning.

The **Writing process** is both a key concept in the teaching of writing and an important research concept in the field of composition studies.

Research on the writing process (sometime called the composing process) focuses on how writers draft, revise, and edit texts. Composing process research was pioneered by scholars such as Janet Emig in The Composing Processes of Twelfth Graders (1971)[1], Sondra Perl in "The Composing Processes of Unskilled College Writers (1979)[2], and Linda Flower and John R. Hayes in "A Cognitive Process Theory of Writing" (1981).[3]

The rest of this page will focus on the writing process as a term used in teaching. In 1972, Donald M. Murray published a brief manifesto on "Teach Writing as a Process Not Product,"[4] a phrase which became a rallying cry for many writing teachers. Ten years later, in 1982, Maxine Hairston argued that the teaching of writing had undergone a "paradigm shift" in moving from a focus on written products to writing processes[5].

Generally the writing process is seen as consisting of five stages:

- Prewriting
- Drafting (See Draft document)
- Revising (See Revision (writing))
- Editing: proofreading
- Publishing

These stages can be described at increasing levels of complexity for both younger students and more advanced writers. The five stages, however, are seldom described as fixed steps in a straightforward process. Rather, they tend to be viewed as overlapping parts of a complex whole. Thus, for instance, a writer might find that, while editing a text, she needs to go back to draft more prose, or to revise earlier parts of what she has written.

Contents

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[edit] Editing

Editing is the stage in the writing process where the writer makes changes in the text to correct errors (spelling, grammar, or mechanics) and fine-tune his or her style. Having revised the draft for content, the writer's task is now to make changes that will improve the actual communication with the reader. Depending on the genre, the writer may choose to adhere to the conventions of Standard English. These conventions are still being developed and the rulings on controversial issues may vary depending on the source. A source like Strunk and White's Elements of Style, first published in 1918, is a well-established authority on stylistic conventions[6]. A more recent handbook for students is Diana Hacker's A Writer's Reference[7]. An electronic resource is the Purdue Online Writing Lab (OWL), where writers may search a specific issue to find an explanation of grammatical and mechanical conventions[8].

Proofread for

- Spelling
- Subject/verb agreement
- Verb tense consistency
- Point of view consistency
- Mechanical errors
- Word choice
- Word usage (there, their or they're)

[9]

Prewriting is the first stage of the writing process, typically followed by drafting, revision, editing and publishing.[1][2][3] Elements of prewriting may include planning, research, outlining, diagramming, storyboarding or clustering (for a technique similar to clustering, see mindmapping).

Contents

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[edit] Motivation and audience awareness

Prewriting usually begins with motivation and audience awareness: what is the student or writer trying to communicate, why is it important to communicate it well and who is the audience for this communication. Writers usually begin with a clear idea of audience, content and the importance of their communication; sometimes, one of these needs to be clarified for the best communication[4][5][6]. Student writers find motivation especially difficult because they are writing for a teacher or for a grade, instead of a real audience.[7] Often teachers try to find a real audience for students by asking them to read to younger classes or to parents, by posting writing for others to read, by writing a blog, or by writing on real topics, such as a letter to the editor of a local newspaper.

[edit] Choosing a topic

One important task in prewriting is choosing a topic and then narrowing it to a length that can be covered in the space allowed.[8] Oral storytelling is an effective way to search for a good topic for a personal narrative. Writers can quickly tell a story and judge from the listeners' reactions whether it will be an interesting topic to write about.

Another way to find a topic is to freewrite, a method first popularized by Peter Elbow. When freewriting, you write any and every idea that comes to mind. This could also be a written exploration of your current knowledge of a broad topic, with the idea that you are looking for a narrow topic to write about. Often freewriting is timed. The writer is instructed to keep writing until the time period ends, which encourages him/her to keep writing past the pre-conceived ideas and hopefully find a more interesting topic.

[edit] Gathering information

Several other methods of choosing a topic overlap with another broad concern of prewriting, that of researching or gathering information. Reading (process) [9] is effective in both choosing and narrowing a topic and in gathering information to include in the writing. As a writer reads other works, it expands ideas, opens possibilities and points toward options for topics and narrowing of topics. It also provides specific content for the eventual writing. One traditional method of tracking the content read is to create annotated note cards with one chunk of information per card. Writers also need to document music, photos, web sites, interviews, and any other source used to prevent plagiarism.

Besides reading what others have written, writers can also make original observations relating to a topic. This requires on-site visits, experimentation with something, or finding original or primary historical documents. Writers interact with the setting or materials and make observations about their experience. For strong writing, particular attention should be given to sensory details (what the writer hears, tastes, touches, smells and feels). While gathering material, often writers pay particular attention to the vocabulary used in discussing the topic. This would include slang, specific terminology, translations of terms, and typical phrases used. The writer often looks up definitions, synonyms and finds ways that different people use the terminology. Lists, journals, teacher-student conference, drawing illustrations, using imagination, restating a problem in multiple ways, watching videos, inventorying interests[10] – these are some of the other methods for gathering information.

[edit] Discussing information

After reading and observing, often writers need to discuss material. They might brainstorm with a group or topics or how to narrow a topic. Or, they might discuss events, ideas, and interpretations with just one other person. Oral storytelling might enter again, as the writer turns it into a narrative, or just tries out ways of using the new terminology. Sometimes writers draw or use information as basis for artwork as a way to understand the material better.[11][12]

[edit] Narrowing the topic

Narrowing a topic is an important step of prewriting. For example, a personal narrative of five pages could be narrowed to an incident that occurred in a thirty minute time period. This restricted time period means the writer must slow down and tell the event moment by moment with many details. By contrast, a five page essay about a three day trip would only skim the surface of the experience. The writer must consider again the goals of communication – content, audience, importance of information – but add to this a consideration of the format for the writing. He or she should consider how much space is allowed for the communication and how What can be effectively communicated within that space?[13]

[edit] Organizing content

At this point, the writer needs to consider the organization of content. Outlining in a hierarchical structure is one of the typical strategies, and usually includes three or more levels in the hierarchy. Typical outlines are organized by chronology, spatial relationships, or by subtopics. Other outlines might include sequences along a continuum: big to little, old to new, etc. Clustering, a technique of creating a visual web that represents associations among ideas, is another help in creating structure, because it reveals relationships. Storyboarding is a method of drawing rough sketches to plan a picture book, a movie script, a graphic novel or other fiction.[14]

[edit] Developmental acquisition of organizing skills

While information on the developmental sequence of organizing skills is sketchy, anecdotal information suggests that children follow this rough sequence: 1) sort into categories[15], 2) structure the categories into a specific order for best communication, using criteria such as which item will best work to catch readers attention in the opening, 3) within a category, sequence information into a specific order for best communication, using criteria such as what will best persuade an audience. At each level, it is important that student writers discuss their decisions; they should understand that categories for a certain topic could be structured in several different ways, all correct. A final skill acquired is the ability to omit information that is not needed in order to communicate effectively.

Even sketchier is information on what types of organization are acquired first, but anecdotal information and research[16] suggests that even young children understand chronological information, making narratives the easiest type of student writing. Persuasive writing usually requires logical thinking and studies in child development indicate that logical thinking is not present until a child is 10–12 years old, making it one of the later writing skills to acquire. Before this age, persuasive writing will rely mostly on emotional arguments.

[edit] Writing trials

Writers also use the prewriting phase to experiment with ways of expressing ideas. For oral storytelling, a writer could tell a story three times, but each time begin at a different time, include or exclude information, end at a different time or place. Writers often try writing the same info. but using different voices, in search of the best way to communicate this information or tell this story.[17]

[edit] Recursion

Prewriting is recursive, that is, it can occur at any time in the writing process and can return several times. For example, after a first draft, a writer may need to return to an information gathering stage, or may need to discuss the material with someone, or may need to adjust the outline. While the writing process is discussed as having distinct stages, in reality, they often overlap and circle back on one another.

[edit] Variables

Prewriting varies depending on the writing task or rhetorical mode. Fiction requires more imagination, while informational essays or expository writing require stronger organization. Persuasive writing must consider not just the information to be communicated, but how best to change the reader's ideas or convictions. Folktales will require extensive reading of the genre to learn common conventions. Each writing task will require a different selection of prewriting strategies, used in a different order.

[edit] Technology

Technological tools are often used in prewriting tasks[18][19][20], including word processors, spreadsheets[21] and publishing programs; however, technology appears to be more useful in the revision, editing and publishing phases of prewriting.

[edit] Writing tests

Teaching writing as a process is accepted pedagogical practice, but there is increasing concern that writing tests do not allow for the full writing process, especially cutting short the time[22][23] needed for prewriting tasks[24].

Revision is the stage in the writing process where the author reviews, alters, and amends her or his message, according to what has been written in the draft. Revision follows drafting and precedes editing. Drafting and revising often form a loop as a work moves back and forth between the two stages. It is not uncommon for professional writers to go through many drafts and revisions before successfully creating an essay that is ready for the next stage: editing.

In their seminal book, *The Elements of Style*, William Strunk, Jr. and E.B. White acknowledge the need for revision in the writing process: "Few writers are so expert that they can produce what they are after on the first try. Quite often you will discover, on examining the completed work, that there are serious flaws in the arrangement of the material, calling for transpositions... do not be afraid to experiment with your text."

Successful revision involves:

Identification of thesis. The purpose of the essay should be re-considered based on what has been written in the draft. If this purpose differs from the original thesis, the author must decide from which thesis to continue writing.

Consideration of structure. The author should identify the strengths of the draft, then re-consider the order of those strengths, adjusting their placement as necessary so the work can build with auxesis to a crescendo.

Uncovering weakness in argument or presentation. Once the strengths of the draft have been identified and placed in the strongest order, the author can re-examine the work for weaknesses in argument or presentation. Faulty logic, missing transitions, and unsupported or poorly supported assertions are common weaknesses. Identifying these weaknesses during revision will inform the next draft.

Successful revision is not improving grammar or diction. Those will be the focus of later editing.

Language education is the teaching and learning of a language. It can include improving a learner's mastery of her or his native language, but the term is more commonly used with regard to second language acquisition, which means the learning of a foreign or second language and which is the topic of this article. Some scholars differentiate between acquisition and learning. Language education is a branch of applied linguistics.

Need for language education

People need to learn a second language because of globalization, connections are becoming inevitable among nations, states and organizations which creates a huge need for knowing another language or more multilingualism. The uses of common languages are in areas such as trade, tourism international relations between governments, technology, media and science. Therefore, many countries such as Japan (Kubota, 1998) and China (Kirkpatrick & Zhichang, 2002) create education policies to teach at least one foreign language at primary and secondary school level. However, some countries such as India, Singapore, Malaysia and Philippines use a second official language in their governing system. According to GAO (2010) many Chinese people are giving enormous importance to foreign language learning, especially learning English Language.

Methods of teaching foreign languages

From Wikipedia, the free encyclopedia
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Main article: Language education

Language education may take place as a general school subject or in a specialized **language school**. There are many methods of teaching languages. Some have fallen into relative obscurity and others are widely used; still others have a small following, but offer useful insights.

While sometimes confused, the terms "approach", "method" and "technique" are hierarchical concepts. An approach is a set of correlative assumptions about the nature of language and language learning, but does not involve procedure or provide any details about how such assumptions should translate into the classroom setting. Such can be related to second language acquisition theory.

There are **three principal views** at this level:

1. The structural view treats language as a system of structurally related elements to code meaning (e.g. grammar).
2. The functional view sees language as a vehicle to express or accomplish a certain function, such as requesting something.
3. The interactive view sees language as a vehicle for the creation and maintenance of social relations, focusing on patterns of moves, acts, negotiation and interaction found in conversational exchanges. This view has been fairly dominant since the 1980s.[1]

A method is a plan for presenting the language material to be learned and should be based upon a selected approach. In order for an approach to be translated into a method, an instructional system must be designed considering the objectives of the teaching/learning, how the content is to be selected and organized, the types of tasks to be performed, the roles of students and the roles of teachers. A technique is a very specific, concrete stratagem or trick designed to accomplish an immediate objective. Such are derived from the controlling method, and less-directly, with the approach.[1]

Structural methods

[edit] Grammar-translation method

Main article: Grammar translation

The grammar translation method instructs students in grammar, and provides vocabulary with direct translations to memorize. It was the predominant method in Europe in the 19th century. Most instructors now acknowledge that this method is ineffective by itself.[citation needed] It is now most commonly used in the traditional instruction of the classical languages, however it remains the most commonly practiced method of English teaching in Japan.[citation needed]

At school, the teaching of grammar consists of a process of training in the rules of a language which must make it possible to all the students to correctly express their opinion, to understand the remarks which are addressed to them and to analyze the texts which they read. The objective is that by the time they leave college, the pupil controls the tools of the language which are the vocabulary, grammar and the orthography, to be able to read, understand and write texts in various contexts. The teaching of grammar examines texts, and develops awareness that language constitutes a system which can be analyzed. This knowledge is acquired gradually, by traversing the facts of language and the syntactic mechanisms, going from simplest to the most complex. The exercises according to the program of the course must untiringly be practiced to allow the assimilation of the rules stated in the course.[citation needed] That supposes that the teacher corrects the exercises. The pupil can follow his progress in practicing the language by comparing his results. Thus can he adapt the grammatical rules and control little by little the internal logic of the syntactic system. The grammatical analysis of sentences constitutes the objective of the teaching of grammar at the school. Its practice makes it possible to recognize a text as a coherent whole and conditions the training of a foreign language. Grammatical terminology serves this objective. Grammar makes it possible for each one to understand how the mother tongue functions, in order to give him the capacity to communicate its thought.

[edit] Audio-lingual method

Main article: Audio-lingual method

The audio-lingual method was developed in the USA around World War II when governments realized that they needed more people who could conduct conversations fluently in a variety of languages, work as interpreters, code-room assistants, and translators. However, since foreign language instruction in that country was heavily focused on reading instruction, no textbooks, other materials or courses existed at the time, so new methods and materials had to be devised. For example, the U.S. Army Specialized Training Program created intensive programs based on the techniques Leonard Bloomfield and other linguists devised for Native American languages, where students interacted intensively with native speakers and a linguist in guided conversations designed to decode its basic grammar and learn the vocabulary. This "informant method" had great success with its small class sizes and motivated learners.[1]

The U.S. Army Specialized Training Program only lasted a few years, but it gained a lot of attention from the popular press and the academic community. Charles Fries set up the first English Language Institute at the University of Michigan, to train English as a second or foreign language teachers. Similar programs were created later at Georgetown University, University of Texas among others based on the methods and techniques used by the military. The developing method had much in common with the British oral approach although the two developed independently. The main difference was the developing audio-lingual methods allegiance to structural linguistics, focusing on grammar and contrastive analysis to find differences between the student's native language and the target language in order to prepare specific materials to address potential problems. These materials strongly emphasized drill as a way to avoid or eliminate these problems.[1]

This first version of the method was originally called the oral method, the aural-oral method or the structural approach. The audio-lingual method truly began to take shape near the end of the 1950s, this time due government pressure resulting from the space race. Courses and techniques were redesigned to add insights from behaviorist psychology to the structural linguistics and constructive analysis already being used. Under this method, students listen to or view recordings of language models acting in situations. Students practice with a variety of drills, and the instructor emphasizes the use of the target language at all times. The idea is that by reinforcing 'correct' behaviors, students will make them into habits.[1]

The typical structure of a chapter employing the Audio-Lingual-Method (ALM—and there was even a text book entitled **ALM** [1963]) was usually standardized as follows: 1. First item was a dialog in the foreign language (FL) to be memorized by the student. The teacher would go over it the day before. 2. There were then questions in the FL about the dialog to be answered by the student(s) in the target language. 3. Often a brief introduction to the grammar of the chapter was next, including the verb(s) and conjugations. 4. The mainstay of the chapter was "pattern practice," which were drills expecting "automatic" responses from the student(s) as a noun, verb conjugation, or agreeing adjective was to be inserted in the blank in the text (or during the teacher's pause). The teacher could have the student use the book or not use it, relative to how homework was assigned. Depending on time, the class could respond as a chorus, or the teacher could pick individuals to respond. It was really a sort of "mimicry-memorization." And it was "Julian Dakin in 'The Language Laboratory and Language Learning' (Longman 1973) who coined the phrase 'meaningless drills' to describe pattern practice of the kind inspired by the above ideas." 5. There was a vocabulary list, sometimes with translations to the mother tongue. 6. The chapter usually ended with a short reading exercise.

Due to weaknesses in performance,[2] and more importantly because of Noam Chomsky's theoretical attack on language learning as a set of habits, audio-lingual methods are rarely the primary method of instruction today. However, elements of the method still survive in many textbooks.[1]

[edit] Proprioceptive language learning method

Main article: Proprioceptive language learning method

The proprioceptive language learning method (commonly called the Feedback training method) emphasizes simultaneous development of cognitive, motor, neurological, and hearing as all part of a comprehensive language learning process. Lesson development is as concerned with the training of the motor and neurological functions of speech as it is with cognitive (memory) functions. It further emphasizes that training of each part of the speech process must be simultaneous. The proprioceptive method, therefore, emphasizes spoken language training, and is primarily used by those wanting to perfect their speaking ability in a target language.

The proprioceptive method bases its methodology on a speech pathology model. It stresses that mere vocabulary and grammar memory is not the sole requirement for spoken language fluency, but that the mind receives real-time feedback from both hearing and neurological receptors of the mouth and related organs in order to constantly regulate the store of vocabulary and grammar memory in the mind during speech.

For optimum effectiveness, it maintains that each of the components of second language acquisition must be encountered simultaneously. It therefore advocates that all memory functions, all motor functions and their neurological receptors, and all feedback from both the mouth and ears must occur at exactly the same moment in time of the instruction. Thus, according to the proprioceptive method, "all student participation must be done at full speaking volume". Further, in order to train memory, after initial acquaintance with the sentences being repeated, "all verbal language drills must be done as a response to the narrated sentences which the student must repeat (or answer) entirely apart from reading a text".[3]

[edit] Functional methods

[edit] The oral approach / situational language teaching

The oral approach was developed from the 1930s to the 1960s by British applied linguists such as Harold Palmer and A.S. Hornsby. They were familiar with the Direct method as well as the work of 19th century applied linguists such as Otto Jesperson and Daniel Jones but attempted to formally develop a scientifically-founded approach to teaching English than was evidenced by the Direct Method.[1]

A number of large-scale investigations about language learning and the increased emphasis on reading skills in the 1920s led to the notion of "vocabulary control". It was discovered that languages have a core basic vocabulary of about 2,000 words that occurred frequently in written texts, and it was assumed that mastery of these would greatly aid reading comprehension. Parallel to this was the notion of "grammar control", emphasizing the sentence patterns most-commonly found in spoken conversation. Such patterns were incorporated into dictionaries and handbooks for students. The principle difference between the oral approach and the direct method was that methods devised under this approach would have theoretical principles guiding the selection of content, gradation of difficulty of exercises and the presentation of such material and exercises. The main proposed benefit was that such theoretically-based organization of content would result in a less-confusing sequence of learning events with better contextualization of the vocabulary and grammatical patterns presented.[1] Last but not least, all language points were to be presented in "situations". Emphasis on this point led to the approach's second name. Proponent claim that this approach leads to students' acquiring good habits to be repeated in their corresponding situations. Teaching methods stress PPP (presentation (introduction of new material in context), practice (a controlled practice phase) and production (activities designed for less-controlled practice)).[1]

Although this approach is all but unknown among language teachers today, elements of it have had long lasting effects on language teaching, being the basis of many widely-used English as a Second/Foreign Language textbooks as late as the 1980s and elements of it still appear in current texts.[1] Many of the structural elements of this approach were called into question in the 1960s, causing modifications of this method that lead to Communicative language teaching. However, its emphasis on oral practice, grammar and sentence patterns still finds widespread support among language teachers and remains popular in countries where foreign language syllabuses are still heavily based on grammar.[1]

[edit] Directed practice

Directed practice has students repeat phrases. This method is used by U.S. diplomatic courses. It can quickly provide a phrasebook-type knowledge of the language. Within these limits, the student's usage is accurate and precise. However the student's choice of what to say is not flexible.

[edit] Interactive methods

[edit] The direct method

Main article: Direct method (education)

The direct method, sometimes also called *natural method*, is a method that refrains from using the learners' native language and just uses the target language. It was established in Germany and France around 1900 and are best represented by the methods devised by Berlitz and de Sauzé although neither claim originality and has been re-invented under other names.[4] The direct method operates on the idea that second language learning must be an imitation of first language learning, as this is the natural way humans learn any language - a child never relies on another language to learn its first language, and thus the mother tongue is not necessary to learn a foreign language. This method places great stress on correct pronunciation and the target language from outset. It advocates teaching of oral skills at the expense of every traditional aim of language teaching. Such methods rely on directly representing an experience into a linguistic construct rather than relying on abstractions like mimicry, translation and memorizing grammar rules and vocabulary.[4]

According to this method, printed language and text must be kept away from second language learner for as long as possible, just as a first language learner does not use printed word until he has good grasp of speech. Learning of writing and spelling should be delayed until after the printed word has been introduced, and grammar and translation should also be avoided because this would involve the application of the learner's first language. All above items must be avoided because they hinder the acquisition of a good oral proficiency.

The method relies on a step-by-step progression based on question-and-answer sessions which begin with naming common objects such as doors, pencils, floors, etc. It provides a motivating start as the learner begins using a foreign language almost immediately. Lessons progress to verb forms and other grammatical structures with the goal of learning about thirty new words per lesson.[4]

[edit] The series method

In the 19th century, Francois Gouin went to Hamburg to learn German. Based on his experience as a Latin teacher, he thought the best way to do this would be memorize a German grammar book and a table of its 248 irregular verbs. However, when he went to the academy to test his new language skills, he was disappointed to find out that he could not understand anything. Trying again, he similarly memorized the 800 root words of the language as well as re-memorizing the grammar and verb forms. However, the results were the same. During this time, he had isolated himself from people around him, so he tried to learn by listening, imitating and conversing with the Germans around him, but found that his carefully-constructed sentences often caused native German speakers to laugh. Again he tried a more classical approach, translation, and even memorizing the entire dictionary but had no better luck.[4]

When he returned home, he found that his three-year-old nephew had learned to speak French. He noticed the boy was very curious and upon his first visit to a mill, he wanted to see everything and be told the name of everything. After digesting the experience silently, he then reenacted his experiences in play, talking about what he learned to whoever would listen or to himself. Gouin decided that language learning was a matter of transforming perceptions into conceptions, using language to represent what one experiences. Language is not an arbitrary set of conventions but a way of thinking and representing the world to oneself. It is not a conditioning process, but one in which the learner actively organizes his perceptions into linguistics concepts.[4]

The series method is a variety of the direct method in that experiences are directly connected to the target language. Gouin felt that such direct "translation" of experience into words, makes for a "living language". (p59) Gouin also noticed that children organize concepts in succession of time, relating a sequence of concepts in the same order. Gouin suggested that students learn a language more quickly and retain it better if it is presented through a chronological sequence of events. Students learn sentences based on an action such as leaving a house in the order in which such would be performed. Gouin found that if the series of sentences are shuffled, their memorization becomes nearly impossible. For this, Gouin preceded psycholinguistic theory of the 20th century. He found that people will memorize events in a logical sequence, even if they are not presented in that order. He also discovered a second insight into memory called "incubation". Linguistic concepts take time to settle in the memory. The learner must use the new concepts frequently after presentation, either by thinking or by speaking, in order to master them. His last crucial observation was that language was learned in sentences with the verb as the most crucial component. Gouin would write a series in two columns: one with the complete sentences and the other with only the verb. With only the verb elements visible, he would have students recite the sequence of actions in full sentences of no more than twenty-five sentences. Another exercise involved having the teacher solicit a sequence of sentences by basically ask him/her what s/he would do next. While Gouin believed that language was rule-governed, he did not believe it should be explicitly taught.[4]

His course was organized on elements of human society and the natural world. He estimated that a language could be learned with 800 to 900 hours of instruction over a series of 4000 exercises and no homework. The idea was that each of the exercises would force the student to think about the vocabulary in terms of its relationship with the natural world. While there is evidence that the method can work extremely well, it has some serious flaws. One of which is the teaching of subjective language, where the students must make judgments about what is experienced in the world (e.g. "bad" and "good") as such do not relate easily to one single common experience. However, the real weakness is that the method is entirely based on one experience of a three-year-old. Gouin did not observe the child's earlier language development such as naming (where only nouns are learned) or the role that stories have in human language development. What distinguishes the series method from the direct method is that vocabulary must be learned by translation from the native language, at least in the beginning.[4]

[edit] Communicative language teaching

Main article: Communicative language teaching

Communicative language teaching(CLT), also known as *the Communicative Approach*, emphasizes interaction as both the means and the ultimate goal of learning a language. Despite a number of criticisms[5] it continues to be popular, particularly in Europe, where constructivist views on language learning and education in general dominate academic discourse. Although the 'Communicative Language Teaching' is not so much a method on its own as it is an approach.[6]

In recent years, task-based language learning (TBLL), also known as task-based language teaching (TBLT) or task-based instruction (TBI), has grown steadily in popularity. TBLL is a further refinement of the CLT approach, emphasizing the successful completion of tasks as both the organizing feature and the basis for assessment of language instruction. Dogme language teaching id s vstriant of TBL.[7] Dogme is a communicative approach, and encourages teaching without published textbooks and instead focusing on conversational communication among the learners and the teacher.[8]

[edit] Language immersion

Main article: Language immersion

Language immersion in school contexts delivers academic content through the medium of a foreign language, providing support for L2 learning and first language maintenance. There are three main types of immersion education programs in the United States: foreign language immersion, dual immersion, and indigenous immersion.

Foreign language immersion programs in the U.S. are designed for students whose home language is English. In the early immersion model, for all or part of the school day elementary school children receive their content (academic) instruction through the medium of another language: Spanish, French, German, Chinese, Japanese, etc. In early total immersion models, children receive all the regular kindergarten and first grade content through the medium of the immersion language; English reading is introduced later, often in the second grade. Most content (math, science, social studies, art, music) continues to be taught through the immersion language. In early partial immersion models, part of the school day (usually 50%) delivers content through the immersion language, and part delivers it through English. French-language immersion programs are common in Canada in the provincial school systems, as part of the drive towards bilingualism and are increasing in number in the United States in public school systems (Curtain & Dahlbert, 2004). Branaman & Rhodes (1998) report that between 1987-1997 the percentage of elementary programs offering foreign language education in the U.S. through immersion grew from 2% to 8% and Curtain & Dahlberg (2004) report 278 foreign language immersion programs in 29 states. Research by Swain and others (Genesee 1987) demonstrate much higher levels of proficiency achieved by children in foreign language immersion programs than in traditional foreign language education elementary school models.

Dual immersion programs in the U.S. are designed for students whose home language is English as well as for students whose home language is the immersion language (usually Spanish). The goal is bilingual students with mastery of both English and the immersion language. As in partial foreign language immersion academic content is delivered through the medium of the immersion language for part of the school day, and through English the rest of the school day.

Indigenous immersion programs in the U.S. are designed for American Indian communities desiring to maintain the use of the native language by delivering elementary school content through the medium of that language. Hawaiian Immersion programs are the largest and most successful in this category.

[edit] Silent Way

The Silent Way[9] is a discovery learning approach, invented by Caleb Gattegno in the 1950s. The teacher is usually silent, leaving room for the students to explore the language. They are responsible for their own learning and are encouraged to interact. The role of the teacher is to give clues, not to model the language.

[edit] Suggestopedia

Main article: Suggestopedia

Suggestopedia was a method that experienced popularity especially in past years, with both staunch supporters and very strong critics, some claiming it is based on pseudoscience.

[edit] Natural Approach

The natural approach is a language teaching method deleoped by Stephen Krashen and Tracy D. Terrell. They emphasise the learner receiving large amounts of comprehensible input. The Natural Approach can be categorized as part of the comprehension approach to language teaching.

[edit] Total Physical Response

Main article: Total Physical Response

In Total Physical Response (TPR), the instructor gives the students commands in the target language and the students act those commands out using whole-body responses. This can be categorized as part of the comprehension approach to language teaching.

[edit] Teaching Proficiency through Reading and Storytelling

Main article: Teaching Proficiency through Reading and Storytelling

Teaching Proficiency through Reading and Storytelling (TPR Storytelling or TPRS) was developed by Blaine Ray, a language teacher in California, in the 1990s. At first it was an offshoot of Total Physical Response that also included storytelling, but it has evolved into a method in its own right and has gained a large following among teachers, particularly in the United States. TPR Storytelling can be categorized as part of the comprehension approach to language teaching.

[edit] Dogme language teaching

Main article: Dogme language teaching

Dogme language teaching is considered to be both a methodology and a movement. Dogme is a communicative approach to language teaching and encourages teaching without published textbooks and instead focusing on conversational communication among the learners and the teacher. It has its roots in an article by the language education author, Scott Thornbury. The Dogme approach is also referred to as “Dogme ELT”, which reflects its origins in the ELT (English language teaching) sector. Although Dogme language teaching gained its name from an analogy with the Dogme 95 film movement (initiated by Lars von Trier), the connection is not considered close.

[edit] Proprietary methods

The following methods are tied to a particular company or school, and are not used in mainstream teaching.

[edit] Pimsleur method

Main article: Pimsleur language learning system

Pimsleur language learning system is based on the research of and model programs developed by American language teacher Paul Pimsleur. It involves recorded 30-minute lessons to be done daily, with each lesson typically featuring a dialog, revision, and new material. Students are asked to translate phrases into the target language, and occasionally to respond in the target language to lines spoken in the target language. The instruction starts in the student's language but gradually changes to the target language. Several all-audio programs now exist to teach various languages using the Pimsleur Method. The syllabus is the same in all languages.

[edit] Michel Thomas Method

Main article: Michel Thomas Method

Michel Thomas Method is an audio-based teaching system developed by Michel Thomas, a language teacher in the USA. It was originally done in person, although since his death it is done via recorded lessons. The instruction is done entirely in the student's own language, although the student's responses are always expected to be in the target language. The method focuses on constructing long sentences with correct grammar and building student confidence. There is no listening practice, and there is no reading or writing. The syllabus is ordered around the easiest and most useful features of the language, and as such is different for each language.[10]

[edit] Other

Appropedia is increasingly being used to as a method to enable service learning in language education.[11][12][13]

There is a lot of language learning software using the multimedia capabilities of computers.

[edit] Learning by teaching (LdL)

Main article: Learning by teaching

Learning by teaching is a widespread method in Germany, developed by Jean-Pol Martin. The students take the teacher's role and teach their peers.

Bilingual education involves teaching academic content in two languages, in a native and secondary language with varying amounts of each language used in accordance with the program model. The following are several different types of bilingual education program models:

- Transitional Bilingual Education. This involves education in a child's native language, typically for no more than three years, to ensure that students do not fall behind in content areas like math, science, and social studies while they are learning English. The goal is to help students transition to mainstream, English-only classrooms as quickly as possible, and the linguistic goal of such programs is English acquisition only.
- **Two-Way** or **Dual Language Immersion** Bilingual Education. These programs are designed to help native and non-native English speakers become bilingual and biliterate. Ideally in such programs in a U.S. context, half of the students will be native speakers of English and half of the students will be native speakers of a minority language such as Spanish. Dual Language programs are less common in US schools, although research indicates they are extremely effective in helping students learn English well and aiding the long-term performance of English learners in school. Native English speakers benefit by learning a second language. English language learners (ELLs) are not segregated from their peers.[1]
- Another form of Bilingual Education is a type of **Dual Language** program that has students study in two different ways: 1) A variety of academic subjects are taught in the students' second language, with specially trained bilingual teachers who can understand students when they ask questions in their native language, but always answer in the second language; and 2) Native language literacy classes improve students' writing and higher-order language skills in their first language. Research has shown that many of the skills learned in the native language can be transferred easily to the second language later. In this type of program, the native language classes do not teach academic subjects. The second-language classes are content-based, rather than grammar-based, so students learn all of their academic subjects in the second language.[*citation needed*]
- **Late-Exit** or **Developmental** Bilingual Education. Education is in the child's native language for an extended duration, accompanied by education in English. The goal is to develop literacy in the child's native language first, and transfer these skills to the second language.

Oral reading strategies

In spite of a common perception that oral reading is an elementary school strategy, it has many benefits for middle grades students—especially those who have not developed strong academic literacy. However, oral reading may be daunting to many young adolescents who are going through an awkward phase in their physical development. They shy away from anything that draws attention to them. As a result, oral reading strategies must be fun, and teachers must ensure that students are safe from taunting or heckling if they do not pronounce words correctly.

The following oral reading strategies capitalize on adolescent needs for relationship building and fun. These strategies also increase oral speaking skills, which are a part of academic literacy.

1. **Readers' theater** encourages students to create plays about material they are learning and to present the play in class. Students get to hear how others use inflection and pacing to convey emotion. The teacher uses the presentation to clarify misconceptions and to make connections between the play and the standards-based lesson.

2. **Think-pair-share** usually pairs a fluent reader with one who needs help. Students take turns reading to each other and share what they have read so they reinforce comprehension.

3. **Popcorn reading** keeps students focused since they do not know when their turn will "pop." In this strategy, one student reads part of a selection. Another "pops in" to continue until the next name is called. This strategy helps content area teachers cover text material in class but does not ensure that the student comprehends the material. It is still the teacher's responsibility to develop metacognitive thinking and comprehension. Through pondering, discussion, and re-reading, students develop comprehension.

4. **Literature circles** are groups of four to six students who read and discuss a novel or article. Each member of a circle takes a turn guiding the group discussion and receiving practice in leadership, group interaction, "argumentative literacy," and responsibility. The circles also allow students to control their own learning and to discuss ideas and concerns about issues raised by the passage. The University of Seattle Web page on Literature Circles (Outside Source) provides many resources to help teachers and includes a link to specific strategies for middle schools.

5. **Guided reading** typically involves the whole class in reading a passage together. It allows the teacher to expose children to a wide range of literature while teaching vocabulary and comprehension strategies.

Researchers suggest that cross-curricular connections help to give students the background knowledge they need to make reading meaningful. Researchers caution against narrowing the curriculum when teachers try to help students improve their reading skills. "Although necessary, being able to read all of the words may not be sufficient because comprehending a text requires other abilities such as knowing the meanings of words, possessing relevant world knowledge, and being able to remember the text already read. Thus, word-reading skill is one of several factors influencing comprehension."[7]

Humanistic "theories" of learning tend to be highly value-driven and hence more like **pre**scriptions (about what ought to happen) rather than **de**scriptions (of what does happen).

- They emphasise the "natural desire" of everyone to learn. Whether this natural desire is to learn whatever it is you are teaching, however, is not clear.
- It follows from this, they maintain, that learners need to be empowered and to have control over the learning process.
- So the teacher relinquishes a great deal of authority and becomes a facilitator.

The school is particularly associated with

- Carl Rogers, and
- Abraham Maslow (psychologists),
- John Holt (child education) and
- Malcolm Knowles (adult education and proponent of andragogy).
- Insofar as he emphasises experiential learning, one could also include Kolb among the humanists as well as the cognitive theorists.

While the tenor of humanistic theory is generally wishy-washy liberal, its approach also underlies the more committed stance of "transformative learning" (Mezirow) and "conscientization" (Freire).

Read more: Humanistic approaches to learning
http://www.learningandteaching.info/learning/humanist.htm#ixzz1DZRmyTSu

Humanistic education is an alternative approach to education based on the work of humanistic psychologists, most notably Abraham Maslow, who developed a famous hierarchy of needs, Carl Rogers, previous president of the American Psychology Association and Rudolf Steiner, the founder of Waldorf education.[1] In humanistic education, the whole person, not just the intellect, is engaged in the growth and development that are the signs of real learning. The emotions, the social being, the mind, and the skills needed for a career direction are all focuses of humanistic education. "Much of a humanist teacher's effort would be put into developing a child's self-esteem. It would be important for children to feel good about themselves (high self-esteem), and to feel that they can set and achieve appropriate goals (high self-efficacy)." [2]

Principles of Humanistic Education

[edit] Choice or Control

The humanistic approach focuses a great deal on student choice and control over the course of their education. Students are encouraged to make choices that range from day-to-day activities to periodically setting future life goals. This allows for students to focus on a specific subject of interest for any amount of time they choose, within reason. Humanistic teachers believe it is important for students to be motivated and engaged in the material they are learning, and this happens when the topic is something the students need and want to know.

[edit] Felt Concern

Humanistic education tends to focus on the felt concerns and interests of the students intertwining with the intellect. It is believed that the overall mood and feeling of the students can either hinder or foster the process of learning.

[edit] The Whole Person

Humanistic educators believe that both feelings and knowledge are important to the learning process. Unlike traditional educators, humanistic teachers do not separate the cognitive and affective domains. This aspect also relates to the curriculum in the sense that lessons and activities provided focus on various aspects of the student and not just rote memorization through note taking and lecturing.

[edit] Self Evaluation

Humanistic educators believe that grades are irrelevant and that only self-evaluation is meaningful. Grading encourages students to work for a grade and not for intrinsic satisfaction. Humanistic educators disagree with routine testing because they teach students rote memorization as opposed to meaningful learning. They also believe testing doesn't provide sufficient educational feedback to the teacher.

[edit] Teacher as a Facilitator

"The tutor or lecturer tends to be more supportive than critical, more understanding than judgmental, more genuine than playing a role." [3] Their job is to foster a engaging environment for the students and ask inquiry based questions that promote meaningful learning.

[edit] Environment

The environment in a school which focuses their practice on humanistic education tends to have a very different setting than a traditional school. It consist of both indoor and outdoor environments with a majority of time being spent outdoors. The indoor setting may contain a few tables and chairs, bean bags for quiet reading and relaxation, book shelf's, hide-aways, kitchens, lots of color and art posted on the walls. The outdoor environment is very engaging for students. You might find tree houses, outdoor kitchens, sand boxes, play sets, natural materials, sporting activities etc. The wide range of activities are offered for students allowing for free choices of interest.

Teaching methods can best be defined as the types of principles and methods used for instruction. There are many types of teaching methods, depending on what information or skill the teacher is trying to convey. Class participation, demonstration, recitation, and memorization are some of the teaching methods being used. When a teacher is deciding on their method, they need to be flexible and willing to adjust their style according to their students. Student success in the classroom is largely based on effective teaching methods.

Diversity in Teaching in the Classroom

For effective teaching to take place, a good method must be adopted by a teacher. A teacher has many options when choosing a style by which to teach. The teacher may write lesson plans of their own, borrow plans from other teachers, or search online or within books for lesson plans. When deciding what **teaching method** to use, a teacher needs to consider students' background knowledge, environment, and learning goals. Teachers are aware that students learn in different ways, but almost all children will respond well to praise. Students have different ways of absorbing information and of demonstrating their knowledge. Teachers often use techniques which cater to multiple learning styles to help students retain information and strengthen understanding. A variety of strategies and methods are used to ensure that all students have equal opportunities to learn. A lesson plan may be carried out in several ways: Questioning, explaining, modeling, collaborating, and demonstrating.

A teaching method that includes questioning is similar to testing. A teacher may ask a series of questions to collect information of what students have learned and what needs to be taught. Testing is another application of questioning. A teacher tests the student on what was previously taught in order to identify if a student has learned the material. Standardized testing is in about every middle school (i.e. Ohio Graduation Test (OGT), Proficiency Test, College entrance Tests (ACT and SAT).

Learning can be done in three ways- Auditory, Visual, and Kinesthetic. It is important to try and include all three as much as possible into your lessons.

[edit] Explaining

This form is similar to lecturing. Lecturing is teaching by giving a discourse on a specific subject that is open to the public, usually given in the classroom. This can also be associated with modeling. Modeling is used as a visual aid to learning. Students can visualize an object or problem, then use reasoning and hypothesizing to determine an answer.

In your lecture you have the opportunity to tackle two types of learning. Not only can explaining (lecture) help the auditory learner through the speech of the teacher, but if the teacher is to include visuals in the form of overheads or slide shows, his/her lecture can have duality. Although a student might only profit substantially from one form of teaching, all students profit some from the different types of learning.

[edit] Demonstrating

Demonstrations are done to provide an opportunity to learn new exploration and visual learning tasks from a different perspective. A teacher may use experimentation to demonstrate ideas in a science class. A demonstration may be used in the circumstance of proving conclusively a fact, as by reasoning or showing evidence.

The uses of storytelling and examples have long since become standard practice in the realm of textual explanation. But while a more narrative style of information presentation is clearly a preferred practice in writing, judging by its' prolificacy, this practice sometimes becomes one of the more ignored aspects of lecture. Lectures, especially in a collegiate environment, often become a setting more geared towards factorial presentation than a setting for narrative and/or connective learning. The use of examples and storytelling likely allows for better understanding but also greater individual ability to relate to the information presented. Learning a list of facts provides a detached and impersonal experience while the same list, containing examples and stories, becomes, potentially, personally relatable. Furthermore, storytelling in information presentation may also reinforce memory retention because it provides connections between factorial presentation and real-world examples/personable experience, thus, putting things into a clearer perspective and allowing for increased neural representation in the brain. Therefore, it is important to provide personable, supplementary, examples in all forms of information presentation because this practice likely allows for greater interest in the subject matter and better information-retention rates.

Often in lecture numbers or stats are used to explain a subject but often when many numbers are being used it is difficult to see the whole picture. Visuals that are bright in color, etc. offer a way for the students to put into perspective the numbers or stats that are being used. If the student can not only hear but see what is being taught, it is more likely they will believe and fully grasp what is being taught. It allows another way for the student to relate to the material.

[edit] Collaborating

Having students work in groups is another way a teacher can direct a lesson. Collaborating allows students to talk with each other and listen to all points of view in the discussion. It helps students think in a less personally biased way. When this lesson plan is carried out, the teacher may be trying to assess the lesson by looking at the student's: ability to work as a team, leadership skills, or presentation abilities. It is one of the direct instructional methods.

A different kind of group work is the discussion. After some preparation and with clearly defined roles as well as interesting topics, discussions may well take up most of the lesson, with the teacher only giving short feedback at the end or even in the following lesson. Discussions can take a variety of forms, e.g. fishbowl discussions.

Collaborating (kinesthetic) is great in that it allows to actively participate in the learning process. These students who learn best this way by being able to relate to the lesson in that they are physically taking part of it in some way. Group projects and discussions are a great way to welcome this type of learning.

[edit] Learning by teaching

Main article: Learning by teaching

Learning by teaching (German:LdL) is a widespread method in Germany, developed by Jean-Pol Martin. The students take the teacher's role and teach their peers.

This method is very effective when done correctly. Having students teach sections of the class as a group or as individuals is a great way to get the students to really study out the topic and understand it so as to teach it to their peers. By having them participate in the teaching process it also builds self-confidence, self-efficacy, and strengthens students speaking and communication skills. Students will not only learn their given topic, but they will gain experience that could be very valuable for life.

[edit] Evolution of teaching methods

[edit] Ancient education

About 3000 BC, with the advent of writing, education became more conscious or self-reflecting, with specialized occupations requiring particular skills and knowledge on how to be a scribe, an astronomer, etc.

Philosophy in ancient Greece led to questions of educational method entering national discourse. In his *Republic*, Plato describes a system of instruction that he felt would lead to an ideal state. In his *Dialogues*, Plato describes the Socratic method.

It has been the intent of many educators since then, such as the Roman educator Quintilian, to find specific, interesting ways to encourage students to use their intelligence and to help them to learn.

[edit] Medieval education

Comenius, in Bohemia, wanted all boys and girls to learn. In his *The World in Pictures*, he gave the first vivid, illustrated textbook which contained much that children would be familiar with in everyday life, and use it to teach the academic subjects they needed to know. Rabelais described how the student Gargantua learned about the world, and what is in it.

Much later, Jean-Jacques Rousseau in his *Emile*, presented methodology to teach children the elements of science and much more. In it, he famously eschewed books, saying the world is one's book. And so Emile was brought out into the woods without breakfast to learn the cardinal directions and the positions of the sun as he found his way home for something to eat.

There was also Johann Heinrich Pestalozzi of Switzerland, whose methodology during Napoleonic warfare enabled refugee children, of a class believed to be unteachable, to learn - and love to learn. He describes this in his account of the educational experiment at Stanz. He felt the key to have children learn is for them to be loved, but his method, though transmitted later in the school for educators he founded, has been thought "too unclear to be taught today". One result was, when he would ask, "Children, do you want to learn more or go to sleep?" they would reply, "Learn more!"

[edit] 19th century - compulsory education

Main article: Prussian education system

The Prussian education system was a system of mandatory education dating to the early 19th century. Parts of the Prussian education system have served as models for the education systems in a number of other countries, including Japan and the United States. The Prussian model had a side effect of requiring additional classroom management skills to be incorporated into the teaching process. [1]

[edit] 20th century

In the 20th century, the philosopher, Eli Siegel, who believed that all children are equally capable of learning regardless of ethnic background or social class, stated: "The purpose of all education is to like the world through knowing it." This is a goal which is implicit in previous educators, but in this principle, it is made conscious. With this principle at basis, teachers, predominantly in New York, have found that students learn the curriculum with the kind of eagerness that Pestalozzi describes for his students at Stanz centuries earlier.

Many current teaching philosophies are aimed at fulfilling the precepts of a curriculum based on Specially Designed Academic Instruction in English (SDAIE). Arguably the qualities of a SDAIE curriculum are as effective if not more so for all 'regular' classrooms.

Some critical ideas in today's education environment include:

- Instructional scaffolding
- Graphic organizers
- Standardized testing

According to Dr. Shaikh Imran, the teaching methodology in education is a new concept in the teaching learning process. New methods involved in the teaching learning process are television, radio, computer, etc.

Other educators believe that the use of technology, while facilitating learning to some degree, is not a substitute for educational method that brings out critical thinking and a desire to learn. Another modern teaching method is inquiry learning and the related inquiry-based science.

Elvis H. Bostwick recently concluded Dr. Cherry's quantitative study "The Interdisciplinary Effect of Hands On Science", a three-year study of 3920 middle school students and their Tennessee State Achievement scores in Math, Science, Reading and Social Studies. Metropolitan Nashville Public School is considered urban demographically and can be compared to many of urban schools nationally and internationally. This study divided students on the basis of whether they had hands on trained teachers over the three-year period addressed by the study.

Students who had a hands-on trained science teacher for one or more years had statistically higher standardized test scores in science, math and social studies. For each additional year of being taught by a hands-on trained teacher, the student's grades increased.

Dyadic interaction (pair work) is a common convention of the language classroom. Some personalities have always worked better than others however. Linguists have found four different types of dyadic interaction:

- collaborative
- dominant/dominant
- dominant/passive
- and expert/novice

The collaborative pattern was found to be the predominant pattern. Both the collaborative dyads and the expert/novice dyads generally engaged in the co-construction of knowledge about language. This knowledge was subsequently appropriated and internalized by members of the dyad. A collaborative dialogue was characterized by both participants being actively involved via requests, explanations, repetitions, suggestions, and repairs.

Any dyads with dominant personality types had the lowest rate of uptake and collaboration. It is common for dominant personality types to be assigned to expert/novice dyads.

Language Output in Pairs

Learning is though to occur when an individual interacts with an interlocutor within his or her zone of proximal development (ZPD) -- that is, in a situation in which the learner is capable of performing at a higher level because there is support from an interlocutor. This is why expert/novice dyads can work so well.

Language output promotes "noticing" -- in producing the target language learners may notice a gap between what they want to say and what they can say. This may trigger cognitive processes which might generate linguistic knowledge that is new for learners.

Language output also allows for hypothesis testing -- producing output tests a language hypothesis about comprehensibility or linguistic well-formedness. Once again, the collaborative dyads allows both learners to have an environment were language output can be easily formed. Dominant dyads would simply avoid specific speech acts as part of their systematic second language performance.

In formal education, a **curriculum** (pronounced /kəˈrɪkjʉləm/; plural: **curricula**, /kəˈrɪkjʉlə/ or **curriculums**) is the set of courses, and their content, offered at a school or university. As an idea, **curriculum** stems from the Latin word for *race course*, referring to the course of deeds and experiences through which children grow to become mature adults. A curriculum is prescriptive, and is based on a more general syllabus which merely specifies what topics must be understood and to what level to achieve a particular grade or standard.

Curriculum in formal schooling

In formal education or schooling (cf. education), a **curriculum** is the set of courses, course work, and content offered at a school or university. A **curriculum** may be partly or entirely determined by an external, authoritative body (i.e. the National **Curriculum** for England in English schools). In the U.S., each state, with the individual school districts, establishes the curricula taught[4]. Each state, however, builds its **curriculum** with great participation of national[5] academic subject groups selected by the United States Department of Education, e.g. National Council of Teachers of Mathematics (NCTM) for mathematical instruction. In Australia each state's Education Department establishes curricula with plans for a National **Curriculum** in 2011. UNESCO's International Bureau of Education has the primary mission of studying curricula and their implementation worldwide.

Curriculum[6] means two things: (i) the range of courses from which students choose what subject matters to study, and (ii) a specific learning program. In the latter case, the **curriculum** collectively describes the teaching, learning, and assessment materials available for a given course of study.

Currently, a ***spiral*** ***curriculum*** is promoted as allowing students to revisit a subject matter's content at the different levels of development of the subject matter being studied. The constructivist approach, of the *tycoil* ***curriculum***, proposes that children learn best via active engagement with the educational environment, i.e. discovery learning. Crucial to the **curriculum** is the definition of the course objectives that usually are expressed as *learning outcomes'* and normally include the program's **assessment strategy**. These outcomes and assessments are grouped as **units** (or modules), and, therefore, the **curriculum** comprises a collection of such units, each, in turn, comprising a specialised, specific part of the **curriculum**. So, a typical **curriculum** includes communications, numeracy, information technology, and social skills units, with specific, specialized teaching of each.

Research can be defined as the search for knowledge, or as any systematic investigation, with an open mind, to establish novel facts, usually using a scientific method. The primary purpose for applied research (as opposed to basic research) is discovering, interpreting, and the development of methods and systems for the advancement of human knowledge on a wide variety of scientific matters of our world and the universe.

Scientific research relies on the application of the scientific method, a harnessing of curiosity. This research provides scientific information and theories for the explanation of the nature and the properties of the world around us. It makes practical applications possible. Scientific research is funded by public authorities, by charitable organizations and by private groups, including many companies. Scientific research can be subdivided into different classifications according to their academic and application disciplines.

Artistic research, also seen as 'practice-based research', can take form when creative works are considered both the research and the object of research itself. It is the debatable body of thought which offers an alternative to purely scientific methods in research in its search for knowledge and truth.

Historical research is embodied in the scientific method.

The phrase *my research* is also used loosely to describe a person's entire collection of information about a particular subject.

Research processes

[edit] Scientific research

Main article: Scientific method

Generally, research is understood to follow a certain structural process. Though step order may vary depending on the subject matter and researcher, the following steps are usually part of most formal research, both basic and applied:

1. Observations and Formation of the topic
2. Hypothesis
3. Conceptual definitions
4. Operational definition
5. Gathering of data
6. Analysis of data
7. Test, revising of hypothesis
8. Conclusion, iteration if necessary

A common misunderstanding is that by this method a hypothesis could be proven or tested. Generally a hypothesis is used to make predictions that can be tested by observing the outcome of an experiment. If the outcome is inconsistent with the hypothesis, then the hypothesis is rejected. However, if the outcome is consistent with the hypothesis, the experiment is said to support the hypothesis. This careful language is used because researchers recognize that alternative hypotheses may also be consistent with the observations. In this sense, a hypothesis can never be proven, but rather only supported by surviving rounds of scientific testing and, eventually, becoming widely thought of as true. A useful hypothesis allows prediction and within the accuracy of observation of the time, the prediction will be verified. As the accuracy of observation improves with time, the hypothesis may no longer provide an accurate prediction. In this case a new hypothesis will arise to challenge the old, and to the extent that the new hypothesis makes more accurate predictions than the old, the new will supplant it.

[edit] Artistic research

One of the characteristics of artistic research is that it must accept subjectivity as opposed to the classical scientific methods. As such, it is similar to the social sciences in using qualitative research and intersubjectivity as tools to apply measurement and critical analysis.[citation needed]

[edit] Historical method

Main article: Historical method

The historical method comprises the techniques and guidelines by which historians use historical sources and other evidence to research and then to write history. There are various history guidelines commonly used by historians in their work, under the headings of external criticism, internal criticism, and synthesis. This includes higher criticism and textual criticism. Though items may vary depending on the subject matter and researcher, the following concepts are usually part of most formal historical research:

- Identification of origin date
- Evidence of localization
- Recognition of authorship
- Analysis of data
- Identification of integrity
- Attribution of credibility

[edit] Research methods

The goal of the research process is to produce new knowledge. This process takes three main forms (although, as previously discussed, the boundaries between them may be obscure.):

- Exploratory research, which structures and identifies new problems
- Constructive research, which develops solutions to a problem
- Empirical research, which tests the feasibility of a solution using empirical evidence

The research room at the New York Public Library, an example of secondary research in progress.

Research can also fall into two distinct types:

- Primary research (collection of data that does not yet exist)
- Secondary research (summary, collation and/or synthesis of existing research)

In social sciences and later in other disciplines, the following two research methods can be applied, depending on the properties of the subject matter and on the objective of the research:

- Qualitative research (understanding of human behavior and the reasons that govern such behavior)
- Quantitative research (systematic empirical investigation of quantitative properties and phenomena and their relationships)

Research is often conducted using the hourglass model Structure of Research.[1] The hourglass model starts with a broad spectrum for research, focusing in on the required information through the methodology of the project (like the neck of the hourglass), then expands the research in the form of discussion and results.

[edit] Publishing

Academic publishing describes a system that is necessary in order for academic scholars to peer review the work and make it available for a wider audience. The 'system', which is probably disorganized enough not to merit the title, varies widely by field, and is also always changing, if often slowly. Most academic work is published in journal article or book form. In publishing, STM publishing is an abbreviation for academic publications in science, technology, and medicine.

Most established academic fields have their own journals and other outlets for publication, though many academic journals are somewhat interdisciplinary, and publish work from several distinct fields or subfields. The kinds of publications that are accepted as contributions of knowledge or research vary greatly between fields; from the print to the electronic format. Business models are different in the electronic environment. Since about the early 1990s, licensing of electronic resources, particularly journals, has been very common. Presently, a major trend, particularly with respect to scholarly journals, is open access. There are two main forms of open access: open access publishing, in which the articles or the whole journal is freely available from the time of publication, and self-archiving, where the author makes a copy of their own work freely available on the web.